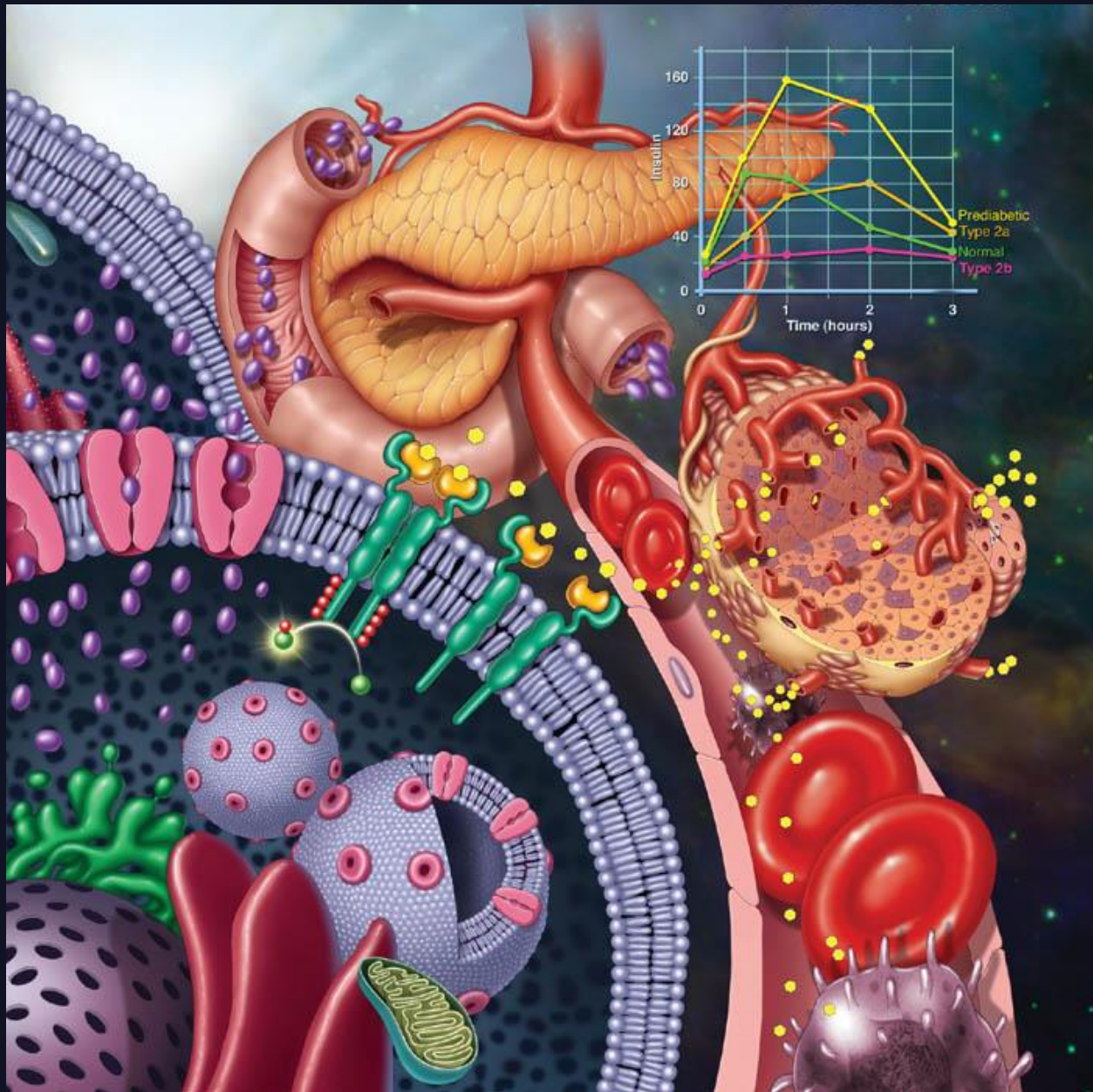




GASTROINTESTINAL MODULE

(NUTRITION, METABOLIC DISORDERS, AND GASTROINTESTINAL SYSTEM)

Study Notes for Exit Exam A

February 2016

“There comes a time when for every addition of knowledge, you forget something that you knew before. It is of the highest importance, therefore, not to have useless facts elbowing out the useful ones.”

—Sir Arthur Conan Doyle, *A Study in Scarlet*

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Embryology

Development and congenital malformations of the Gastrointestinal Tract and accessory glands

Foregut—pharynx to duodenum. Midgut—duodenum to proximal 2/3 of transverse colon. Hindgut—distal 1/3 of transverse colon to anal canal above pectinate line.

Foregut development and rotation:

- The abdominal foregut rotates **90° (clockwise)** around its longitudinal axis.

Midgut development and rotation:

- 6th week—midgut herniates through umbilical cord
- 10th week—returns to abdominal cavity + rotates around superior mesenteric artery (SMA)
- During this herniation into & from the umbilical cord, the midgut undergoes a **270° counterclockwise** rotation around the axis of the **superior mesenteric artery**.

Allantois: An outpouching of the hindgut that is nonfunctional in the human embryo and is obliterated early. It connects to the apex of the developing bladder at the proximal end and travels through the umbilical cord. The proximal portion of the allantois is termed the **urachus**, which spans from the umbilicus to the bladder. Once obliterated, the urachus is termed the **median umbilical ligament**. It continues to have no physiologic use but may be used as a surgical landmark to indicate the midline of the abdomen.

Vitelline Duct (Omphalomesenteric Duct): Tube that connects the embryologic yolk sac to the Mid-gut and provides nourishment to the embryo. It is obliterated in the seventh week but may partially persist in the form of **Meckel diverticulum**.

Cloaca: The terminal portion of the hindgut, and the early embryologic cavity into which the gastrointestinal and genitourinary systems empty. The cloaca will later divide into the rectum and the **urogenital sinus**.

Pancreas and Spleen

Pancreas—derived from foregut. Ventral pancreatic buds contribute to uncinete process and main pancreatic duct. The dorsal pancreatic bud alone becomes the body, tail, isthmus, and accessory pancreatic duct. Both the ventral and dorsal buds contribute to the pancreatic head.

Spleen—arises in mesentery of stomach (hence is mesodermal) but is supplied by foregut (celiac artery).

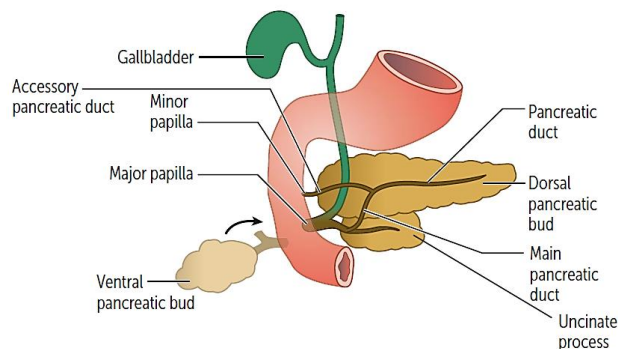


Figure 1: Development of Pancreas

Liver

The hepatic diverticulum develops as an outgrowth of the endoderm of the foregut in the region of the duodenum near the border between the foregut and the midgut.

- This diverticulum enters the ventral embryonic mesentery and distally becomes the liver and gallbladder and proximally becomes the biliary duct system.
- The part of the ventral embryonic mesentery between the liver and gut tube becomes the lesser omentum, and the part between the liver and ventral body wall becomes the falciform ligament.

Development and Congenital Anomalies of the Diaphragm

Diaphragm is a musculotendinous structure sitting between two cavities: thoracic and abdominal. It is made up of four different parts:

- **Septum transversum:** primordium for the central tendon of the diaphragm; fuses with pleuroperitoneal membrane and dorsal mesentery of the esophagus;
- **Pleuroperitoneal membrane: primordial diaphragm**
- Dorsal mesentery of esophagus: constitutes medial portion of the diaphragm; **crura of the diaphragm** which attach to body of vertebrae
- Muscular ingrowth from the lateral body walls: During the 9th to 12th weeks, the lungs and pleural cavities enlarge, "burrowing" into the lateral body walls. During this process, the body-wall tissue is split into two layers:
 - An external layer that becomes part of the definitive abdominal wall
 - An **internal layer** that contributes to **peripheral parts of the diaphragm**, external to the parts derived from the pleuroperitoneal membranes

Positional Changes and Innervation of the Diaphragm

Septum transversum lies in the neck region opposite to 3-5th cervical somites before relocation of the heart. During the fifth week, myoblasts from these somites migrate into the developing diaphragm, bringing their nerve fibers with them [from 3-5th ventral rami forming **phrenic nerves**].

As the dorsal part of embryo is apparently growing faster, phrenic nerves start to extend with descending diaphragm [6th- thoracic somites level and beginning of 8th week at 1st lumbar level]. Phrenic nerve passes through pleuroperitoneal membrane [=fibrous pericardium – adult derivative] to enter all parts of diaphragm except periphery- derived from lateral walls---thus supplied by lower intercostal nerves.

Congenital malformations of diaphragm

Congenital diaphragmatic hernia

It is condition in which abdominal contents are herniated in to pleural cavity. It's caused due to failure of pleuroperitoneal membrane to **develop** or **fuse** with **other components**. It commonly occurs on **left posterolateral side** – life threatening due its compression on lung buds causing **pulmonary hypoplasia**. It is characterized by the **unusually flat abdomen**, breathlessness and severe dyspnea. It can also cause polyhydramnios (=excess amniotic fluid). It is killing so needs immediate surgical repair.

Foregut	Midgut	Hindgut
Artery: celiac	Artery: superior mesenteric	Artery: inferior mesenteric
Parasympathetic innervation: vagus nerves	Parasympathetic innervation: vagus nerves	Parasympathetic innervation: pelvic splanchnic nerves
Sympathetic innervation: - Preganglionics: thoracic splanchnic nerves, T5–T9 - Postganglionic cell bodies: celiac ganglion	Sympathetic innervation: - Preganglionics: thoracic splanchnic nerves, T9–T12 - Postganglionic cell bodies: superior mesenteric ganglion	Sympathetic innervation: - Preganglionics: lumbar splanchnic nerves, L1–L2 - Postganglionic cell bodies: inferior mesenteric ganglion
Referred Pain: Epigastrium	Referred Pain: Umbilical	Referred Pain: Hypogastrium
Foregut Derivatives	Midgut Derivatives	Hindgut Derivatives
Esophagus Stomach Duodenum (first and second parts) Liver Pancreas Biliary apparatus Gallbladder	Duodenum (second, third, and fourth parts) Jejunum Ileum Cecum Appendix Ascending colon Transverse colon (proximal two-thirds)	Transverse colon (distal third—splenic flexure) Descending colon Sigmoid colon Rectum Anal canal (above pectinate line)

Table 1: Embryologic divisions of Gastrointestinal system

Important Congenital Malformations of the Gastrointestinal System	
Malformation	Clinical Features
Hypertrophic Pyloric Stenosis	Thickening of the pylorus musculature Projectile vomiting Palpable knot “olive” in the pyloric region
Extrahepatic biliary atresia	Incomplete recanalization of the bile duct during development Presents shortly after birth Dark urine; Clay-colored stool Jaundice
Annular pancreas	Abnormal fusion of ventral & dorsal pancreatic buds, forming a constricting ring around duodenum (the 2 nd part) Duodenal obstruction (bilious vomiting); presents shortly after birth
Pancreatic divisum	ventral and dorsal parts fail to fuse at 8 weeks. Common anomaly; mostly asymptomatic, but may cause chronic abdominal pain and/or pancreatitis.
Meckel Diverticulum	Persistent remnant of the vitelline duct Forms an outpouching (true diverticulum) in the ileum Ulceration and bleeding Fifty percent contain either gastric or pancreatic tissue when symptomatic
Malrotation of the midgut	Normal 270-degree rotation is not completed Cecum and appendix lie in upper abdomen Associated with volvulus (twisting of intestine), causing an obstruction (cont'd)

	(cont'd)
Duodenal stenosis or atresia	Results from failure of the normal recanalization of the lumen May produce failure to thrive Trisomy 21 – associated with down's syndrome
Hirschsprung disease (congenital or toxic megacolon)	Failure of neural crest cells to migrate to colon No peristalsis Constipation and abdominal distention in newborn Bowel movement precipitated by digital rectal examination
Anal agenesis	Lack of anal opening as a result of improper formation of urorectal septum May cause rectovesical (anus to bladder), rectovaginal, or rectourethral fistula
¹Gastroschisis	Extrusion of abdominal contents through abdominal folds Not covered by peritoneum
*Omphalocele	Persistence of herniation of abdominal contents into umbilical cord Sealed by peritoneum
Esophageal atresia (EA)	EA with distal tracheoesophageal fistula (TEF) is the most common (85%) Choking and vomiting (with first feed) – b/c of atretic esophagus Reflux related aspiration – b/c of comm. b/n distal esophagus & trachea

Gross Anatomy

It is obvious that tables and notes won't count in anatomy, so please refer Netter's Atlas (the 6th Ed.) for pictures while reading this notes. No major pictures have been included here because of page limit.

Anterior-Lateral Abdominal Wall

Surface Anatomy

1. Recall the basic terminology used to define the surface anatomy of the regions of the abdomen

Depending on the context or clinical situation, there are two different ways that the abdomen can be divided. The "quick and dirty" way is to draw two lines, one down the midline and the other horizontal through the umbilicus. The result is four quadrants: the **right upper quadrant, left upper quadrant, right lower quadrant, and left lower quadrant**. The contents are as follows:

Right Upper Quadrant	rt. lobe of liver, gallbladder , pylorus, the first three parts of the duodenum, head of the pancreas , rt. suprarenal (adrenal) gland, rt. kidney , rt. colic (hepatic) flexure, sup. part of ascending colon, rt. half of transverse colon
Left Upper Quadrant	lt. lobe of liver, spleen, most of stomach , jejunum and proximal ileum, body and tail of pancreas , lt. suprarenal (adrenal) gland, lt. kidney , lt. colic (splenic) flexure, sup. part of descending colon, lt. half of transverse colon
Right Lower Quadrant	cecum, appendix, most of ileum , inf. part of ascending colon, rt. ovary, rt. uterine tube, rt. ureter, rt. spermatic cord
Left Lower Quadrant	sigmoid colon, inf. part of descending colon, lt. ovary, lt. uterine tube, lt. ureter, lt. spermatic cord

The other way is to draw four lines. Two are horizontal, one at the lower rib margins at L3 (subcostal plane) and the other passing through the iliac tubercles at L5 (intertubercular plane). The other two are vertical,

¹ Both omphalocele and Gastroschisis are failure of lateral body fold to fuse; hence are body wall defects (↑ AFP)

passing through the middle of each clavicle (midclavicular line). The result is nine regions (from up to down & right to left): the **right hypochondriac region, epigastric region, left hypochondriac region, right lumbar region, umbilical region, left lumbar region, right inguinal region, hypogastric region, and the left inguinal region**. Do not worry about knowing the *exact* contents of each specific region. Clinically important are epigastric (pain from heartburn, ulcer), umbilical (visceral pain often refers here), right inguinal (appendix), left hypochondriac (spleen tenderness), right hypochondriac (for palpating liver and gallbladder to check for hepatomegaly).

Layers

1. Describe the layers of the anterolateral abdominal wall.

The anterior portion and the lateral portion of the abdominal wall are often considered together (anterolateral) because the muscles are functionally similar, and are functionally *different* than those of the posterior aspect. The anterior wall has peculiarities above & below the arcuate line.

Lateral: from superficial to deep: skin, fatty layer of superficial fascia (Camper's² fascia), membranous layer of superficial fascia (Scarpa's fascia), deep fascia, external abdominal oblique muscle, deep fascia, internal abdominal oblique muscle, deep fascia, transversus abdominis muscle, transversalis fascia, extraperitoneal fat, parietal peritoneum

Anterior (superior to arcuate line): from superficial to deep: skin, fatty layer of superficial fascia, membranous layer of superficial fascia, deep fascia, **external abdominal oblique aponeurosis***, **anterior layer of internal abdominal oblique aponeurosis***, rectus abdominis, **posterior layer of internal abdominal oblique aponeurosis***, **transversus abdominis aponeurosis***, transversalis fascia, extraperitoneal fat, parietal peritoneum

Anterior (inferior to arcuate line): from superficial to deep: skin, fatty layer of superficial fascia, membranous layer of superficial fascia, deep fascia, **external abdominal oblique aponeurosis***, **internal abdominal oblique aponeurosis***, **transversus abdominis aponeurosis***, rectus abdominis, transversalis fascia, extraperitoneal fat, parietal peritoneum

*** rectus sheath**

2. Describe the formation of the rectus sheath

The **rectus sheath** is formed by the interweaving of the **aponeuroses** of the **external oblique, internal oblique, and transversus abdominis muscles**. It encloses the **rectus abdominis and pyramidalis muscles** (only present 80% of the time). It is divided into two parts, superior and inferior, by the **arcuate line**, which is halfway between the **pubic crest** and the **umbilicus**. The superior part of the sheath completely surrounds the rectus abdominis muscles. The **anterior layer (lamina)** is made of contributions from the external abdominal oblique and the internal abdominal oblique aponeuroses. The **posterior layer (lamina)** is composed of fibers from the transversus abdominis aponeurosis and the internal abdominal oblique aponeurosis. (the internal abdominal oblique aponeurosis splits into two parts, one contributing to the anterior and the other contributing to the posterior layers of the sheath.) The inferior part is deficient posteriorly because all three aponeuroses contribute to the anterior layer of the sheath. The only thing left on the posterior

² Mnemonic: **Fatty Campers** tend to be lazy. Fatty Layer: Camper's fascia

aspect of the rectus abdominis muscle, then, is the thin **transversalis fascia**. The fibers all meet in the middle and interweave at the **linea alba** (a vertical fibrous band that extends from the symphysis pubis to the xiphoid process and lies in the midline).

Muscles

1. Describe the muscles of the anterior abdominal wall, their actions and the arrangement of fibers.

Muscle	Action	Notes	Image
external abdominal oblique	flexes and laterally bends the trunk	The fibers of external abdominal oblique course down and medially (hands in pocket); the external spermatic fascia is the external abdominal oblique muscle's contribution to the coverings of the testis and spermatic cord	
internal abdominal oblique	flexes and laterally bends the trunk	The fibers of internal abdominal oblique course up and medially, perpendicular to the fibers of external abdominal oblique; the cremaster muscle and fascia is the internal abdominal oblique muscle's contribution to the coverings of the testis and spermatic cord	
transversus abdominis	flexes and laterally bends trunk	transversus abdominis muscle does not contribute to the coverings of the spermatic cord and testis; transversalis fascia, the deep fascia that covers the inner surface of the transversus abdominis, forms the internal spermatic fascia	
rectus abdominis	flexes the trunk	rectus sheath contains rectus abdominis and is formed by the aponeuroses of external and internal oblique and transversus abdominis mm. (Latin, rectus = straight)	
pyramidalis	draws the linea alba inferiorly	the pyramidalis m. is not always present	
cremaster	elevates testis (not well developed in females)	the cremaster m. is internal abdominal oblique muscle's contribution to the coverings of the spermatic cord and testis; the cremasteric reflex may be elicited by stroking the medial thigh (where the femoral branch of the genitofemoral n. distributes cutaneously) (Greek, cremaster = a suspender)	

Nerves

- thoracoabdominal nerves (branches of the VPR³ of T7-T11):** travel anteroinferiorly between the internal oblique and transverse abdominal muscles (remember the analogous

³ Ventral Primary Rami of spinal nerves

situation in the thorax). Supplies motor (to the muscles) and sensory (cutaneous) fibers. Distribution is as follows:

1. T7-T9 - superior to **umbilicus**
2. T10 - at level of umbilicus
3. T11 (along with subcostal, iliohypogastric, and ilioinguinal nerves) - inferior to umbilicus.
2. **subcostal nerves (T12)**: travel anteroinferiorly between the internal oblique and transverse abdominal muscles (remember the analogous situation in the thorax) to innervate the wall inferior to the umbilicus. Supplies motor (to the muscles) and sensory (cutaneous) fibers.
3. **iliohypogastric nerves (L1)**: path is somewhat similar to thoracoabdominal nerves and subcostal nerves, that is, anteroinferiorly between the internal oblique and transverse abdominal muscles for part of the way. However, the iliohypogastric nerves and ilioinguinal nerves are different in that they pierce the internal abdominal oblique at the anterior superior iliac spine to travel superficial to it and deep to the external abdominal oblique. Supplies motor (to the muscles) and sensory (cutaneous) fibers to the wall inferior to the umbilicus.
4. **ilioinguinal nerves (L1)**: supplies motor (to the muscles) and sensory (cutaneous) fibers to the wall inferior to the umbilicus. Sometimes considered separate from the iliohypogastric nerves because **ilioinguinal** nerves also innervate the **scrotum** or **labia** by sending branches through the **inguinal canal**.

Clinical Anatomy of the abdominal wall

A twenty-five-year-old female medical student presents to the emergency room with a complaint of "colicky" **periumbilical** (around umbilicus) **pain** which has intensified over the last 6-8 hours and now has started to migrate to the **right lower quadrant**. The patient reports some initial nausea, and as the pain has increased she has had increasing **emesis** and **anorexia**. Physical exam demonstrates the patient has no distension, auscultation reveals hyperactive bowel sounds, and on palpation the patient demonstrates **rebound tenderness**, and the **muscles** of the anterior wall in the **right lower quadrant** are rigid. In addition, the patient has a low-grade fever, and laboratory tests reveal a rising white blood cell count. The attending determines that the patient has acute **appendicitis** and prepares to take the student to the O.R. for an appendectomy. The surgeon asks you the following questions regarding the surgery.

1. What signs and symptoms revealed to the physician that the patient had an acute abdominal problem - potentially appendicitis?

Abdominal pain (periumbilical shifting to the right lower quadrant) followed by anorexia, nausea, and vomiting.

2. How would you distinguish between gastroenteritis and appendicitis?

Gastroenteritis usually causes vomiting before pain and is associated with diarrhea. The patient will also present with a less acute history of abdominal discomfort.

3. What is **McBurney's point**?

It is the point within the right lower quadrant which can be defined at: one-third the distance from the anterior iliac spine to the umbilicus on a line connecting the two.

4. What is **McBurney's sign**?

Tenderness at McBurney's point in patients with appendicitis.

5. What types of incisions can be made in the abdominal wall?

Surgeons choose the following based on their ability to provide access to the desired organ:

a. **Median or midline:** Cut through the linea alba, superior or inferior to the umbilicus. Benefit: minimal blood loss, avoids major nerves, and easy access for exploration.

b. **Paramedian incisions:** Cut to the right or left of the midline. Benefit: Avoid nerves, frees the rectus abdominal muscle which decreases tension to the muscle. Gives access to the peritoneal cavity.

c. **McBurney incisions:** Incision of the external oblique aponeurosis in the direction of its fibers. The internal oblique and transversus abdominis are then incised and split in the direction of their fibers which are then retracted. Benefit: Provides good access with almost no muscle damage and it avoids damage to local nerves.

d. **Suprapubic incision:** This transverse, slightly convex cut transects the linea alba and anterior layer of the rectus sheath at the pubic hairline. Separate the underlying rectus muscles via the tendons (to allow better reattachment) and identify the surrounding nerves. Benefits: Use for most gynecologic surgeries.

e. **Transverse incision:** Cuts through the anterior rectus sheath and the rectus abdominis. Benefits: Causes the least amount of nerve damage, the muscular segments can be rejoined, they are incredibly useful for dissection above the level of the umbilicus.

6. Which of these incisions would be the most ideal for an **appendectomy**?

Transverse incisions have become increasingly more popular replacing the traditional McBurney's incision. Transverse incisions are less likely to cause nerve injury. Currently, there are more simple and high tech tools for carrying out appendectomy without an incision.

7. When placing an incision in the abdominal wall, what nerves have to be identified? What would be a consequence of damage to the nerves?

The inferior thoracic spinal nerves (T11 and T12), the iliohypogastric nerve and the ilioinguinal nerve innervate abdominal muscles and skin inferior to the umbilicus

(discussed above). Therefore, damage to these nerves can lead to muscle weakness with a potential for herniation and pain/altered sensation due to both these muscular and cutaneous branches.

8. Why were the muscles of the anterior abdominal wall rigid?

The ruptured appendix caused peritonitis. Inflammation of the parietal peritoneum causes pain and reflex spasm of the abdominal wall over that area – the peritoneum, abdominal muscles, and overlying skin are all supplied by the same segmental nerves (T11, T12, L1). Spasm apparently occurs to immobilize that region of the abdomen in order to minimize spread of the infection.

9. What nerve is at risk when an incision is made at McBurney's point? What would be the long-term effects of damage to this nerve?

Ilioinguinal and iliohypogastric nerves. Damage to the nerve could cause partial paralysis of the internal oblique and transversus abdominis muscles resulting in laxity of the conjoint tendon and possibly inguinal hernia (will be discussed later). Sensation in the anterior scrotal and lateral part of the penis might also be affected.

10. If the appendix is not immediately visible, what features of the cecum could the surgeon use to locate it?

The teniae coli muscles can be traced along the cecum to locate the base of the appendix.

11. Fortunately for the patient, she was treated before her infection became life-threatening; however, if she had waited to seek treatment or if the physicians had not acted quickly on their suspicions, her infection may have continued to spread in the peritoneal cavity and the blood stream. Which organ is particularly at risk for secondary infection in appendicitis and why?

The liver. The vast majority of blood from the GI tract drains into the portal vein and then filters through the liver before entering the inferior vena cava and passing into the general circulation. As a result, the liver is particularly susceptible to secondary infection in appendicitis.

Abdominal Viscera

Surface Anatomy

(surface anatomy of the abdominal organs was more or less discussed with the abdominal wall above).

Relations

1. Identify and describe the parts and peritoneal relations of the duodenum and pancreas

Because of their close functional and spatial relationships, as well as a shared blood supply, it is logical to consider the duodenum and the pancreas together.

The **duodenum** is the first 25cm part of the small intestine connecting the stomach and the small intestine. (FYI: both "du-uh-DEE-num" and "du-AH-de-num" are correct pronunciations.) It is mostly retroperitoneal except the first part (ampulla). The duodenum has four parts:

- **superior (1st): horizontal**, lies at **L1** level. This part is:
 - posterior to the gallbladder and quadrate lobe of the liver;
 - anterior to the bile duct, the **gastroduodenal artery***⁴, the portal vein, and the IVC;
 - inferior to the neck of the gallbladder; and
 - superior to the neck of the pancreas.
- **descending (2nd): vertical**, descends from **L1 to L3**, **point of entry for bile & pancreatic ducts**. This part is:
 - posterior to the transverse colon and mesocolon, and also posterior to part of the jejunum;
 - anterior to the hilum of the right kidney, the renal vessels, **the right ureter**, and the right psoas major muscle; and
 - lateral to the head of the pancreas and pancreatic and bile ducts.
- **horizontal (3rd): horizontal**, crosses **L3** vertebra. This part is:
 - **posterior to the SMA, SMV**, and parts of the jejunum;
 - anterior to the right psoas major muscle, the IVC, **the abdominal aorta**, and the **right ureter**; and
 - inferior to the head of the pancreas (on the right side) and the SMA and SMV.
- **ascending (4th): vertical**, ascends from **L3 to L2**. This part is:
 - posterior to the root of the mesentery, parts of the jejunum;
 - anterior to the left psoas muscle;
 - lateral to the head of the pancreas (on the left side); and
 - inferior to the body of the pancreas.

The **pancreas** is an **accessory** digestive organ. It is retroperitoneal, except for its tail. It makes enzymes that aid in chemical digestion of food. (This is its "exocrine" role. The pancreas also has an "endocrine" role with insulin, glucagon, and all that business.) It is also divided into four parts:

- **head**: attached to the duodenum, includes the uncinate process. The head of the pancreas rests on the IVC, the right renal artery and vein, and the left renal vein.
- **neck**: short junction between the head and the body. It overlies the SMA and SMV, including the junction of the SMV and splenic vein to form the portal vein. It is posterior to the pylorus of the stomach.
- **body**: between the neck and the tail. Posteriorly it contacts the SMA, abdominal aorta, and the left suprarenal (adrenal) gland, kidney, and renal vessels. Anterior to it is the omental bursa.

⁴ This is clinically important. Duodenal ulcers result hemorrhage from the **gastroduodenal artery** (know this relation well). The others, except for red, are mentioned just for the sake of completeness. But knowing corresponding lumbar vertebrae levels might be important. Also know that the **2nd part** is site of **entry for bile & pancreatic ducts**.

- **tail:** the somewhat mobile end of the pancreas. It lies within the splenorenal ligament and contacts the left kidney, left colic (splenic) flexure, and the hilum of the spleen posteriorly.

2. Discuss the pathway of common entry of the bile ducts and pancreatic ducts into the 2nd part of the duodenum.

- Fresh bile from the liver drains out via the **right** and **left hepatic ducts**, which join to become the **common hepatic duct**. Stored bile leaves the **gall bladder**⁵ via the **cystic duct**. (Please note that bile from the liver also *enters* the gallbladder via the cystic duct, by traveling retrograde, when the GI tract does not need bile to aid in digestion of food and bile is therefore stored.) The common hepatic duct joins the cystic duct to form the **common bile duct**.
- The **main pancreatic duct** begins in the tail of the pancreas and receives tributaries all along its path to the head of the pancreas. Often there is also a separate **accessory pancreatic duct**, which drains independently into the duodenum via the lesser (minor) duodenal papilla.
- Usually, the main pancreatic joins the common bile duct right before entering the duodenum, so they both dump their contents via the greater (major) duodenal papilla (comprised of the ampulla of Vater, or hepatopancreatic ampulla, and the sphincter of Oddi). There is variation in this anatomy, for example, independent dumping of the pancreatic and common bile ducts into the duodenum via separate papillae.

3. Identify parts of the liver – the surfaces (& their relations) and mention the lobes of the liver.

Surfaces:

- **diaphragmatic:** the anterior and superior (and some of the posterior) parts of the liver are in contact with the diaphragm.
- **Visceral (inferior):** most of the posterior and inferior surface area contacts abdominal viscera, including the body and pyloric area of the stomach, the superior (1st) part of the duodenum, the lesser omentum, the gallbladder, the right side of the transverse colon (including the right colic (hepatic) flexure), and the right kidney and suprarenal (adrenal) glands.

Anatomical Lobes

The gallbladder, IVC, ligamentum teres, ligamentum venosum, and porta hepatis all combine to form an H-shape on the liver's visceral surface. The anatomical lobes can be described with respect to this group of structures:

1. **right lobe:** to the right of the "H"
2. **left lobe:** to the left of the "H"

⁵ Gallbladder: The **gallbladder** is an organ which lies in a fossa on the visceral surface of the liver to the right of the quadrate lobe. It stores and concentrates bile, which enters and leaves through the cystic duct.

3. **quadrate lobe:** between the gallbladder and the ligamentum teres (the top spines of the "H") ("quadrate" because it's square)
4. **caudate lobe:** between the IVC and the ligamentum venosum (the bottom spines of the "H") ("caudate" because it has a tail)

4. Identify the parts of the stomach and describe its spatial relations to surrounding organs & mesenteries.

Parts:

- **cardia:** immediately surrounding the esophageal opening
- **fundus:** the part that extends to the left and superior to the esophageal opening, filling the dome of the diaphragm. It is separated from the esophageal opening by the **cardiac notch**.
- **body:** between the fundus and the pyloric portion
- **pyloric portion (pylorus):** includes the **pyloric antrum** (the entryway to the pyloric canal), the pyloric canal, and the pylorus *proper*, which thickens to form the **pyloric sphincter**. It is separated from the body by the **angular notch**.

Curvatures:

1. **lesser curvature:** forms the right/superior/concave border of the stomach
2. **greater curvature:** forms the left/inferior/convex border of the stomach

Orientation and relations:

The plane of the stomach is oblique, running from the fundus to the pylorus inferiorly, ventrally, and to the right. It has, then, effectively an anterior and a posterior surface. The left anterior surface contacts the diaphragm and the abdominal wall, while on the right side the anterior surface lies against the liver. The posterior surface of the stomach forms the anterior wall of the omental bursa. The stomach is almost entirely covered in peritoneum, except where vessels run along the curvatures and where the peritoneum reflects to form ligaments, or attachments to other viscera. These include the hepatogastric ligament from the lesser omentum, and the gastrophrenic, gastrosplenic, and gastrocolic ligaments from the greater omentum.

4. Describe the basic organization of the peritoneum and peritoneal cavity, including subdivisions, mesenteries, and ligaments.

The **peritoneum**, like the pericardium and pleura, is a **serous membrane** that invests viscera. It is comprised of **parietal** and **visceral** peritoneum. There are many specializations of the peritoneum.

Mesenteries: result from the invagination of "intraperitoneal" organs into the sac. The mesenteries connect viscera to the **posterior** abdominal wall and are **VERY** important in that they conduct blood vessels and nerves. (There are no vessels *within* the peritoneal cavity, of course.) In contrast

to the intraperitoneal, the **retroperitoneal**⁶ structures lack mesentery and also include non-GI structures. Injury to the retroperitoneal structures can cause blood or gas accumulation in retroperitoneal space. The mesentery of the colon is usually called the "mesocolon". For example, we speak of the "**transverse mesocolon**" and the "**sigmoid mesocolon**". (The other parts of the colon are not completely invested by peritoneum, and are therefore "**retroperitoneal**".) Also, often "**the mesentery**" refers specifically to the mesentery of the small intestine. **Mesoappendix**: is a continuation of "the mesentery" that carries the blood supply for the appendix - the appendicular artery.

Omentum (pl. omenta): generally, refers to a *free* fold of peritoneum. This is exemplified by the **greater omentum**, which attaches to the stomach, droops far down into the abdominal cavity, and comes back up to attach to the transverse colon. The **lesser omentum**, on the other hand, is not really "free". It connects the stomach to the liver, and its membranous portion is called the hepatogastric ligament.

Ligaments: connect organs to one another or to the abdominal wall. Usually, in this case, we are referring to the stomach or the liver.

- Falciform ligament**: between liver and abdominal wall,
- hepatogastric ligament**: between stomach and liver; part of lesser omentum
- hepatoduodenal ligament**: b/n liver and duodenum; part of lesser omentum
- gastrophrenic ligament**: b/n stomach and diaphragm,
- gastrosplenic ligament**: b/n stomach and spleen,
- gastrocolic ligament**: b/n stomach and transverse colon,
- splenorenal ligament**: between the spleen and the left kidney

The peritoneal *cavity*, analogous to the pleural cavity, is a space between the two layers of peritoneum. Fluid in this space provides lubrication, allowing the abdominal viscera to move around. The peritoneal cavity contains no viscera. There are two major subdivisions of the peritoneal cavity:

Greater sac: the main part of the peritoneal cavity. It is divided into two sections. The **supracolic** and **infracolic compartments** are divided by the **transverse mesocolon**. The infracolic compartment is further divided into right and left infracolic spaces by "the mesentery".

Lesser sac (omental bursa): lies posterior to the stomach. It is divided into two sections, a **superior** and an **inferior recess**.

5. Describe the basic anatomy of large & small intestines including blood supply & internal structure.

The **small intestine** comprises the duodenum, jejunum, and ileum. The first and shortest portion of the small intestine is the duodenum, but due to its intimate relationship with the liver and pancreas, we discussed it with those structures. The remainder of the small intestine is divided into

⁶ Mnemonic for retroperitoneal organs – **SAD PUCKER**: **S**uprarenal (adrenal) glands, **A**orta & IVC, **D**uodenum (except 1st part), **P**ancreas (except tail), **U**reters, **C**olon (descending & ascending only; not transverse & sigmoid), **K**idneys, **E**sophagus (thoracic portion only), and **R**ectum (partially).

the **jejunum** and the **ileum**. The border between them is not distinct, but the **jejunum** is usually thought to lie mostly in the left upper quadrant, comprising about 40% of the total length, while the **ileum** lies mostly in the right lower quadrant, providing the remaining **60%**.

The small intestine contains circular folds of tissue that are covered with **villi**. (This becomes important in histology.) Added to the fact that the long tube is coiled in the abdominal cavity, there is **A LOT** of surface area over which nutrients can be absorbed.

The colon (the **large intestine**) is considered in five sections. From the **cecum**, where the small intestine attaches, the **ascending colon** travels up to the right colic (hepatic) flexure. From there, the **transverse colon** moves across to the left colic (splenic) flexure. Then, the **descending colon** travels down, becoming the **sigmoid colon** which ends in the rectum.

The colon, or large intestine, has three features that distinguish it from the small intestine:

- Its longitudinal muscle forms three **teniae coli**, instead of being continuous all the way around.
- The teniae coli are actually shorter than the colon, so viscera forms **haustra coli** (bulges, sacculations).
- The surface of the colon has fat-filled tabs, called **epiploid appendages**

Blood Supply

1. Describe the blood supply of the abdominal foregut via branches of the celiac artery.

There are three branches of the celiac trunk (look the figure on the next page):

1. **left gastric artery**: supplies the lesser curvature of the stomach, anastomoses with the **right gastric artery**
2. **splenic artery**: supplies the spleen. It gives off **left gastro-omental artery**, which supplies the greater curvature of the stomach and anastomoses with the **right gastro-omental artery**.
3. **common hepatic artery**: supplies the liver via the **proper hepatic artery**. The latter gives off the **right gastric artery**, which supplies the lesser curvature of the stomach and anastomoses with the **left gastric artery**. The common hepatic also gives off the **gastroduodenal artery**, which supplies the duodenum, pancreas, and, via the **right gastro-omental artery**, supplies the greater curvature of the stomach.

Looking at things the other way around, the stomach is supplied by:

1. **right gastric artery**: a branch of the **proper hepatic artery**, from the **common hepatic artery**, which is a primary branch of the **celiac trunk**
2. **left gastric artery**: a primary branch of the **celiac trunk**
3. **right gastro-omental artery**: a branch of the **gastroduodenal artery**, which is a branch of the **common hepatic artery**, which is a primary branch of the **celiac trunk**
4. **left gastro-omental artery**: a branch of the **splenic artery**, which is a primary branch of the **celiac trunk**

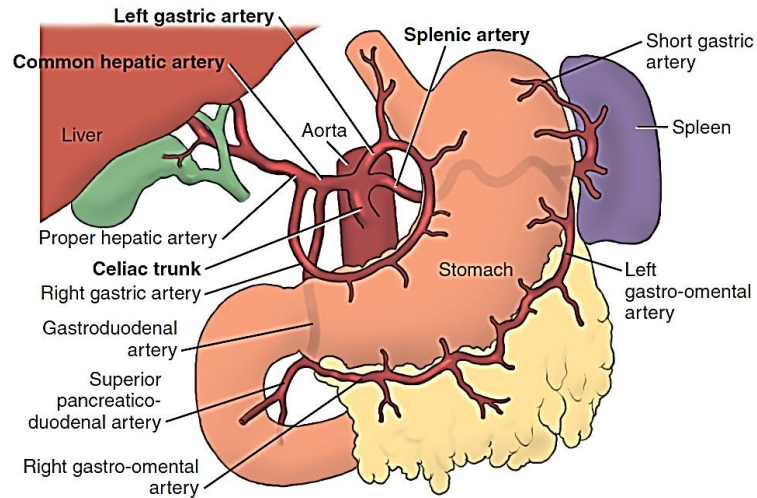


Figure 1: Branches of the celiac trunk. (From Morton DA, Albertine KH, Peterson KD. Gray's Dissection Guide for Human Anatomy. 2nd ed. Philadelphia: Elsevier; 2006.) Gray's Dissection Guide for Human Anatomy. 2nd ed. Philadelphia: Elsevier; 2006.)

2. Describe the pattern of **common** vasculature of the duodenum and pancreas.

Both the duodenum and the pancreas receive blood from the **celiac trunk** and the **superior mesenteric artery**. They serve as a site for anastomosis between these two large systems. The body and tail of the pancreas are supplied mostly by branches of the splenic artery, namely the **great, superior, caudal**, and **dorsal pancreatic arteries**.

3. Describe the relationships of its portal venous, hepatic arterial, and hepatic venous circulation.

The liver receives arterial blood from the aorta via the **celiac trunk**, which gives off the **common hepatic artery** as a branch. This artery gives off the **proper hepatic artery**, which then divides into the **right** and **left hepatic arteries**. Further branching occurs as the arteries supply the hepatic tissue.

The liver also receives the **portal vein**, which brings nutrients and other compounds absorbed from the GI tract to be stored and/or processed. Within the liver, the portal vein divides into right and left branches of the portal vein, then further into segmental branches.

The portal venous and hepatic arterial circulation are closely related structurally, with *very* similar branching patterns within the right lobe. The other lobes have similar branching as well, but are subject to more variation than the right. (The bile ducts also follow this pattern, but of course they are transporting their contents in the other direction.) This total arrangement is similar in concept to the pulmonary circulation, where the pulmonary arteries, bronchial arteries, and bronchioles have similar branching patterns. Continuing that line of thought, the hepatic veins are similar to the pulmonary veins in that they reside between lobes (segments) and drain blood from two or more adjoining areas.

4. Discuss the blood supply to the small and large intestines.

The small intestine is completely supplied by the **superior mesenteric artery (SMA)**, via 15 or so **intestinal (jejunal and ileal) arteries**. These smaller branches eventually turn into **arcades (arterial arches)**, providing **collateral circulation** to different sections of the organ. **Vasa recta (Straight arteries)** then jump from the arcades to the **tissue**.

The large intestine (the colon) receives blood from both the **superior mesenteric artery (SMA)** and the **inferior mesenteric artery (IMA)**. The SMA supplies the **colon up to the splenic flexure**, via the following branches:

- **ileocolic artery:** divides into ileal and colic branches to supply the cecum and the ascending colon. The ileocolic artery also gives off the appendicular artery that supplies the vermiform appendix.
- **right colic artery:** supplies the ascending colon
- **middle colic artery:** supplies the transverse colon

The IMA supplies the rest of the colon via:

- **left colic artery:** supplies the descending colon
- **sigmoid arteries:** 3 or 4 branches that supply the descending and sigmoid colon
- **superior rectal artery:** supplies upper rectum

Innervation

The GI Tract has a rather complex nerve supply, divided into two major groups: extrinsic (parasympathetic and sympathetic innervation), and intrinsic (enteric) innervation.

Extrinsic Innervation

Extrinsic motor and sensory innervation is supplied by both **sympathetic** and **parasympathetic** nerves.

Sympathetic innervation

Arises at spinal cord levels **T5-L1**. The preganglionic fibers synapse with the **prevertebral abdominal ganglia (celiac, superior, and inferior mesenteric)**. The primary preganglionic neurotransmitter is **acetylcholine (ACh)**, which interacts with the **nicotinic receptors** of the postganglionic nerves. The postganglionic fibers then spread to all parts of the gut via **splanchnic nerves** (thoracic and sacral splanchnic nerves) and terminate on the neurons of the enteric nervous system. The primary postganglionic neurotransmitter is **norepinephrine**, which acts as an **inhibitory** neurotransmitter in the GI tract, **decreasing motility** as well as secretory and digestive functions.

Parasympathetic innervation

Supplied by **vagus** and **pelvic nerves**: The **vagus nerve (CN-X)** provides both parasympathetic afferent and efferent innervation to and from the esophagus, stomach,

gallbladder, pancreas, entire small intestine, and the proximal half of the large intestine. Pelvic nerves arising at S2 and S3 innervate the distal half of the large intestine, entire sigmoid colon, and rectum. Preganglionic fibers of the parasympathetic (vagus and sacral) nerves **synapse directly** in the enteric nervous system, releasing **ACh** and activating **nicotinic** receptors. Postganglionic fibers **only** run in the **wall** of the organ they innervate and activate the **muscarinic** receptors (via **ACh**). The parasympathetic system **increases** the **activity** level of the enteric nervous system.

Intrinsic Innervation – also known as the enteric nervous system, or the “third division” of the autonomic nervous system. It is composed of a series of ganglionic nerve plexuses, contained entirely within the gut wall and running from the esophagus to anus. Enteric neurons use many neurotransmitters, most notably, neuropeptides. The two principal components of the enteric nervous system are the **myenteric** (Auerbach’s) and **submucosal** (Meissner’s) nerve plexuses (will be discussed in physiology).

Clinical Anatomy of the Abdominal Visceral Organs

Clinical Anatomy of the Duodenum, Liver, Pancreas and Gallbladder

M.S., a generally healthy 74-year-old woman, visited her physician with the following complaints: **progressive jaundice** over the last week or so, frequent bowel movements with pale, greasy feces, a lack of energy, weight loss, and back pain. The physician ordered a series of tests, which suggested that the **jaundice** was of an **obstructive**, not metabolic, nature. Abdominal ultrasound demonstrated the presence of a growth on the **head of the pancreas**⁷, and further tests indicated that the lesion was a **pancreatic carcinoma**. After consultation with a surgeon, M.S. elected to have a **cholecystojejunostomy** performed to correct the obstruction and prevent the discomfort and **pruritis**⁸ that usually accompany obstructive jaundice. The pancreatic tumor was believed to be inoperable, and M.S. was referred to her local hospital for assistance as her disease progressed. Questions to consider:

1. How does the tumor in the head of the pancreas relate to the **jaundice** experienced by the patient?

Tumors in the head of the pancreas often obstruct the **common bile duct**⁹, resulting in blockade of the normal bile recycling circuit. This blockade prevents excretion of **bilirubin**, a yellow-colored pigment that is a red blood cell breakdown product. The accumulation of bilirubin in various tissues, including the skin, causes jaundice.

2. How might the **pancreatic carcinoma** relate to M.S.'s back pain?

Compression of somatic nerves on the posterior abdominal wall as a **pancreatic carcinoma** expands would likely cause back pain.

⁷ This is the most common site for carcinoma of the pancreas (ductal carcinoma of pancreas).

⁸ Generalized or localized itching due to irritation of sensory nerve endings

⁹ The main pancreatic duct may also be constricted by the growing tumor – this also causes jaundice.

3. Pancreatic carcinomas frequently metastasize to the **liver**, resulting in significant **hepatomegaly**. Why is the liver a common target for metastasis?

Blood-borne metastases from a pancreatic tumor would enter the **portal vein** via either the **splenic** or **superior mesenteric vein**. All blood in the portal system filters through the liver before entering the inferior vena cava.

4. What other organs, due to their proximity to the pancreas, may be susceptible to invasion by a pancreatic carcinoma?

The structures in the immediate vicinity of the head of the pancreas, including duodenum, common bile duct, spleen, suprarenal gland, transverse colon, and stomach are all vulnerable to invasion by a pancreatic carcinoma.

A 56 years old slightly obese female patient with epigastric pain, slight jaundice, and a radiating back pain¹⁰ presents to your hospital. The physician you're working with suspects the patient is suffering from pancreatitis secondary to gallstones. After a thorough examination, the physician informs you that she has a necrotizing pancreatitis.

1. What are some causes of pancreatitis?

- Alcohol, Gallstones (as in this case), Medications - antibiotics (erythromycin, tetracycline), antifungals (metronidazole), H2 blockers, and diuretics, Hyperlipidemia, hypercalcemia, Perforating gastric ulcer, Exotic worms, such as Ascaris, Medical procedures

2. What was the likely primary cause in this case?

Blockage of the **common bile duct** by **gallstones**, which caused a reflux into the pancreas or blocked the **pancreatic duct**.

3. What structures could be affected by this disease process?

Uncinate process, head of the pancreas, bile duct, duodenum, and mesenteric vessels and spleen, as the inflammation spreads via vascular pathways.

4. Describe the blood supply to the head, body and tail of the pancreas:

Head: Supplied by the **gastrooduodenal artery** (from common hepatic branch of celiac trunk), which gives off the **posterior superior** and **anterior superior pancreaticoduodenal arteries**. The **superior mesenteric artery** also supplies the **head** via the **anterior** and **posterior inferior pancreaticoduodenal arteries**.

Body/Tail: The **dorsal pancreatic artery**, a branch of the splenic artery sends a branch to the left

¹⁰ Many GI disorders present with abdominal pain—use the quality of pain to help you: **sharp, stabbing epigastric** pain that radiates to the back suggests pancreatitis; **burning epigastric** pain suggests a gastric or duodenal ulcer; and pain that **shifts** from the epigastrium to the right lower quadrant suggests appendicitis.

that forms the **inferior pancreatic artery**. Branches of the splenic artery and the inferior pancreatic artery in turn provide multiple branches to the tail.

A forty-year-old slightly obese female presents to your E.R. with spasmodic, colicky pain in her right upper quadrant, with some right shoulder discomfort and accompanying nausea and vomiting. She reports that she has been running a fever over the past twenty-four hours. On physical exam you note heart rate 110, blood pressure 110/70, respiratory rate 20, and temperature 102. She is noticeably uncomfortable on deep inspiration, and palpation reveals an increase in pain while palpating the RUQ. The physician suspects that the patient is suffering from an acute episode of **cholecystitis** and orders an ultrasound study; the study reveals a large stone lodged within the **cystic duct**. The patient is prepared for a cholecystectomy. Questions and answers:

1. What is **cholecystitis**, and what are some of its causes?

Cholecystitis means inflammation of the gallbladder. This generally occurs as a result of the formation of **gallstones** that become lodged in a portion of the gallbladder and block passage of bile through the cystic duct.

2. What is the clinical presentation of cholecystitis?

Nausea, vomiting, fever, colicky spasmodic pain to the right upper quadrant which does not tolerate deep palpation, and possibly an enlarged gallbladder.

3. What are the three parts of the gallbladder?

The fundus, body, and neck.

4. Why might the patient experience right shoulder discomfort?

The primary nerves that innervate the gallbladder originate from the celiac plexus (which provides sympathetic innervation), and the vagus nerve (provides parasympathetic). Sensory innervation is provided by the right phrenic nerve (C3, 4, 5). The stimulation of the **right phrenic nerve** is responsible for the referred pain to the right shoulder.

Clinical Anatomy of the Stomach and Spleen

A 42-year-old business executive was admitted to the hospital after visiting the emergency room complaining of severe **epigastric pain** and pain over her right shoulder. She had a history of **gastric ulcer** which had been treated previously with medication, but on questioning, she admitted that she had been so busy recently that she had forgotten to refill her prescription and had not taken her medication in some time. As a result of the history and physical findings, the physician suspected that she was suffering from a perforated gastric ulcer. Gastrosocopy was performed which confirmed the diagnosis. When the surgeon examined the patient's stomach during the surgery, she found a small perforation on the posterior aspect of the body of the stomach near the lesser curvature. The perforation was repaired and, in addition, a **vagotomy** was performed. During the vagotomy, the surgeon found it necessary to cut the **left gastric artery** and ligate it. Questions to consider:

1. What structures are at risk for damage by gastric juices if a perforation like the one described above occurs?

Any structure located posterior to the stomach could be damaged by gastric juices leaking from a perforated ulcer on the posterior wall of the stomach. These organs would include the **pancreas**, the **left suprarenal gland**, the **upper part of the left kidney**, the **diaphragm**, the **splenic artery**, and its branches, and maybe the **spleen**. Erosion of the wall of the **splenic artery** is of particular concern because it could lead to severe internal hemorrhage and rapid blood loss.

2. Why did the patient experience pain over her shoulder as well as in her abdomen?

The shoulder pain was likely referred pain that occurred due to irritation of the diaphragm by gastric juices. The diaphragm is innervated by the **phrenic nerves** which arise from cervical nerves 3, 4, and 5. These cervical nerves also contain nerves that innervate the shoulder region; thus, the shoulder is a common location for referred pain from the diaphragm.

3. What is a vagotomy and why was it performed?

A **vagotomy** involves severing some or all of the branches of the anterior and posterior vagal trunks. In severe cases of gastric ulcer, this procedure is sometimes performed to reduce the acid secretion of the stomach, which is stimulated, in part, by parasympathetic innervation provided by the **vagus nerves**. A partial **gastrectomy**, which involves removing the antrum of the stomach, may also be performed for this purpose.

4. Since the **left gastric artery** had to be ligated during the surgery, how will the stomach obtain an adequate blood supply?

If the left gastric artery is occluded due to surgical ligation or some other event, collateral circulation from the **right gastric artery** (a branch of the proper hepatic artery), **right gastro-omental artery** (a branch of the gastroduodenal artery), and the **left gastro-omental** and **short gastric arteries** (branches of the splenic artery) is usually adequate to compensate for loss of flow through the left gastric artery.

5. Variations in the arteries of the celiac trunk are quite common, and thus are of particular interest to surgeons working in this area. Suppose the **common hepatic artery** originated from the **left gastric artery** in this case (an uncommon, but possible, variation) and the surgeon had to ligate the left gastric artery proximal to the bifurcation. How would this affect blood flow to the stomach? to other organs?

The anatomic variation described in this question is probably quite rare, but if the **common hepatic artery** did originate from the **left gastric artery** and the left gastric were ligated, the only blood flow to the stomach would be through the left gastro-omental artery from the splenic artery (unless there was retrograde flow into the gastroduodenal artery from the superior mesenteric artery). This may not be sufficient for the stomach's survival. In addition, critical blood flow to the liver, gall bladder, duodenum, and pancreas would be interrupted, leading to the death of the patient unless blood flow is promptly restored.

A twenty-six-year old female presents to your Emergency Room after a motor vehicle accident. The patient reports that she was a pedestrian and while crossing the street was struck by a car. The patient's only complaints are abdominal tenderness and left shoulder pain. Her abdomen is tender on palpation of the **left upper quadrant**, with a faint tire mark over that area; in addition, you detect **crepitus** (indicating fractured ribs) over the 9th, 10th, and 11th ribs on the left side. Several tests confirm a complete splenic rupture and fractured ribs. The patient is taken to the O.R. for a **splenectomy**. In the O.R, the surgeon tests your preclinical knowledge by the following questions.

1. What overlying structures protect the spleen?

Ribs 9, 10, and 11 lie over the region of the spleen; the spleen cannot normally be felt on deep palpation below the costal margin, unless significantly enlarged.

2. What other structures lie within this quadrant and are at risk for injury?

Stomach, splenic flexure of the colon, tail of the pancreas, left kidney, and suprarenal gland, as well as the spleen.

3. What is the major arterial supply of the spleen? What other major branches arise from this artery?

The **splenic artery**, a branch of the **celiac trunk**, is the major arterial blood source to the spleen. The branches of the splenic artery include the **left gastro-omental**, and the **short gastric arteries**.

4. What is the major vein that drains the spleen, and what is its course?

The **splenic vein** drains the spleen. It leaves the hilum of the spleen to join the **inferior mesenteric vein**, which then joins the **superior mesenteric vein** to form the **portal vein**.

5. What are the signs and symptoms of acute splenic rupture?

Hypotension, **left upper quadrant pain**, peritoneal irritation, and referred pain to the left shoulder.

Inguinal Region and Inguinal Hernias

1. Describe the anatomy of the inguinal canal. (Greek, Inguinal: groin)

The inguinal canal can be thought of as a tunnel that travels from an "entrance", which is lateral and deep, to an "exit", which is medial and superficial. (As described below, during the development of the gonads there are structures that travel through the canal in this way.) It, like a tunnel, also has a roof, a floor, and two walls.

Contents: **spermatic cord** (in males) or the **round ligament of the uterus** (in females) as well as **blood vessels**, **lymphatic vessels**, and the **ilioinguinal nerve** (which enters the canal from the side, rather than passing through the deep inguinal ring).

Openings:

- ✓ **deep (internal) inguinal ring:** the entrance to the canal. The transversalis fascia pouches out, creating an opening through which structures can leave the abdominal cavity.
- ✓ **superficial (external) inguinal ring:** the exit from the canal. It is formed by the splitting of the diagonal fibers of the external abdominal oblique aponeurosis.

Defined by:

- ✓ **roof:** fibers of internal abdominal oblique and transversus abdominis muscles.
- ✓ **floor: inguinal ligament** throughout, with **lacunar ligament** added medially
- ✓ **anterior wall:** external abdominal oblique aponeurosis throughout, with internal abdominal oblique aponeurosis added laterally
- ✓ **posterior wall:** mostly transversalis fascia, with **conjoint tendon (falx inguinalis)**, which is the joining of internal abdominal oblique and transversus abdominis aponeuroses, medially.

Note: When studying the inguinal region, watching a video/animation is recommended because it can be confusing with all of these layers mixing and matching and making structures. On top of that, it is important clinically!

2. Describe the anatomy of the various kinds of inguinal hernias (indirect and direct inguinal).

- **Indirect:** **congenital**, **lateral** to inferior epigastric vessels, enter canal via **deep inguinal ring**, **often** enter scrotum
- **Direct:** **acquired**, **medial** to inferior epigastric vessels, literally **pierce the canal**, **seldom** enter scrotum

Also, give some thought to what coverings each would have (coming up next).

3. Review the descent of the gonads.

Briefly, the testes, initially retroperitoneal organs, start out behind the peritoneum below the kidney, in the lumbar region. From there, they descend through extraperitoneal connective tissue through the inguinal canal to end in the scrotal cavity. A **gubernaculum** (fibrous tissue) helps to pull the testes down into the scrotal cavity through the **inguinal canal**. In adult life, the gubernaculum will become the **scrotal ligament**. The female case is similar, but the uterus develops near the middle of the gubernaculum and interrupts the descent of the ovary at the pelvic brim. The gubernaculum ends up as a *short ligament* from ovary to uterus (**ovarian ligament**) and a *longer ligament* passing from the uterus (through the inguinal canal, inguinal rings) to the labia majora (**round ligament of the uterus**). Keep in mind that the descent involves an **invagination of the abdominal wall** and that this is where the inguinal canal starts (deep ring), this is also how the spermatic cord gets its layers from the layers of the abdominal wall. The peritoneum initially invaginates through the abdominal wall parallel to the round ligament (female) or spermatic cord (male) before birth. This normally closes but if it remains open, constitutes a congenital inguinal hernia.

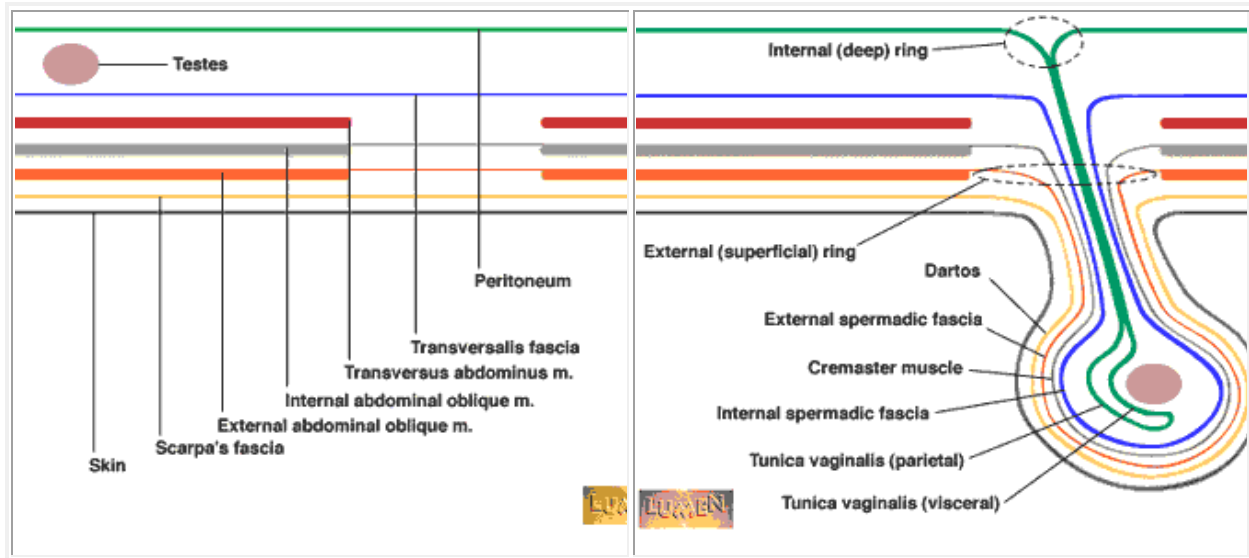


Figure 2: layers of anterior abdominal wall and contributions to the covering of the spermatic cord

These diagrams are from the Loyola University Chicago - website. See an animation of this process [here](#).

3. Define the layers of the anterior abdominal wall, their contributions to the coverings of the spermatic cord and round ligament, and the origin of these coverings as related to the descent of the gonads.

<u>Layer of anterolateral abdominal wall</u>	<u>Spermatic cord/testes covering</u>
skin	scrotum
superficial fascia (only scarpa's fascia)	scrotum/ dartos fascia & muscle
external abdominal oblique aponeurosis	external spermatic fascia
internal abdominal oblique muscle	cremaster muscle
fascia from internal abdominal oblique muscle	cremasteric fascia
transversalis fascia	internal spermatic fascia
peritoneum	processus vaginalis (obliterated)/ tunica vaginalis

4. What tissue forms the deep (internal) inguinal ring, and how does it do so?

The deep inguinal ring is formed by transversalis fascia above the midpoint of the inguinal ligament. It is not really "ring"-like, but is more of a sleeve.

5. What passes through the deep ring?

In the male, the structures that make up the spermatic cord traverse the ring (actually, the cord is formed at the ring). In the female, it is the round ligament of the uterus.

6. Define the inguinal triangle (aka Hesselbach's triangle) using its boundaries.

This is the weak area defined **medially**: by lateral border of **rectus abdominis muscle**, **laterally**: by the **inferior epigastric artery**, and **inferiorly**: by the **inguinal ligament**.

7. What is the relationship between direct and indirect inguinal hernias and the falx inguinalis, weak fascia, deep and superficial inguinal rings, inguinal canal, and inguinal ligament?

OK... here we go AGAIN! (this means it's important to know this) **Direct inguinal hernias** pass through the inguinal triangle, which is an area of weak fascia. They don't start at the deep inguinal ring and almost never go into the scrotum (this is what makes them different from indirect inguinal hernias). Both the superficial inguinal ring and the inherent weakness of abdominal wall lateral to the falx inguinalis make this area susceptible to hernias. **Indirect inguinal hernias** start at deep inguinal ring, pass down inguinal canal, through superficial ring, and, in the male, usually descend into scrotum along with the spermatic cord. In the female they travel along the round ligament.

Clinical Anatomy of Inguinal region: Inguinal Hernia

S.T., a 35-year-old man, was carrying furniture out to a van in preparation for his family's move to a new home. When S.T. strained to pick up a particularly heavy coffee table, he suddenly felt a sharp pain in his right groin. Later, he noticed that a painful bulge had developed in his groin which disappeared when he laid on his back. He did not like going to the doctor, so he ignored the condition. After several months, the pain and the bulge in his groin increased and he finally consented to see a physician. On examination, the physician observed a swelling which began about midway between the anterior superior iliac spine and the midline, progressed medially for about 4 cm, and then turned toward the scrotum. Taking the history and physical findings into account, the physician made a diagnosis of indirect inguinal hernia and scheduled S.T. for surgery. The hernia was successfully repaired, and S.T. was released from the hospital a few days later.

1. What defines this hernia as an indirect inguinal hernia rather than a direct inguinal hernia? List the key features of each.

This hernia is an **indirect inguinal hernia** because it traverses the **inguinal canal**. Indirect inguinal hernial sacs enter the inguinal canal at the deep inguinal ring **lateral** to the **inferior epigastric vessels**. Depending on the severity, the hernial sac may or may not extend through the superficial inguinal ring and into the scrotum or labium majus. Indirect inguinal hernias are more common in males than in females, and may be congenital or the result of an injury, as in the case above. **Direct inguinal hernias** bulge through **the weak fascia** of the abdominal wall **directly behind** the superficial inguinal ring and **medial** to the inferior epigastric vessels. Direct inguinal hernias are common in elderly men with weak abdominal muscles. Direct inguinal hernias rarely enter the scrotum.

2. What caused the bulge? What body layers would surround it as it proceeded into the scrotum and what abdominal layers are they derived from?

The bulge was most likely caused by a **loop** of small intestine that traversed the inguinal canal. The body layers surrounding the intestinal bulge in the scrotum are as follows: skin, dartos muscle, membranous layer of the superficial fascia, external spermatic fascia (from the external oblique aponeurosis), the cremasteric fascia (from the internal oblique aponeurosis), and the internal spermatic fascia (from the transversalis fascia).

3. Why would it be necessary to repair a hernia like the one described above as quickly as possible?

An indirect inguinal hernia in which the bowel becomes entrapped can rapidly lead to an **intestinal obstruction** and to **strangulation** of the loop of bowel projecting into the inguinal canal. If not treated immediately, **gangrene** of the intestine may set in within 12 hours.

4. What are common causes of **direct inguinal hernias**?

Increased intra-abdominal pressure caused by straining, heavy lifting, heavy exercise, obesity, chronic cough. (weight lifters are particularly susceptible to this kind of hernia).

Portal vein, Portal Hypertension and Portosystemic Anastomosis

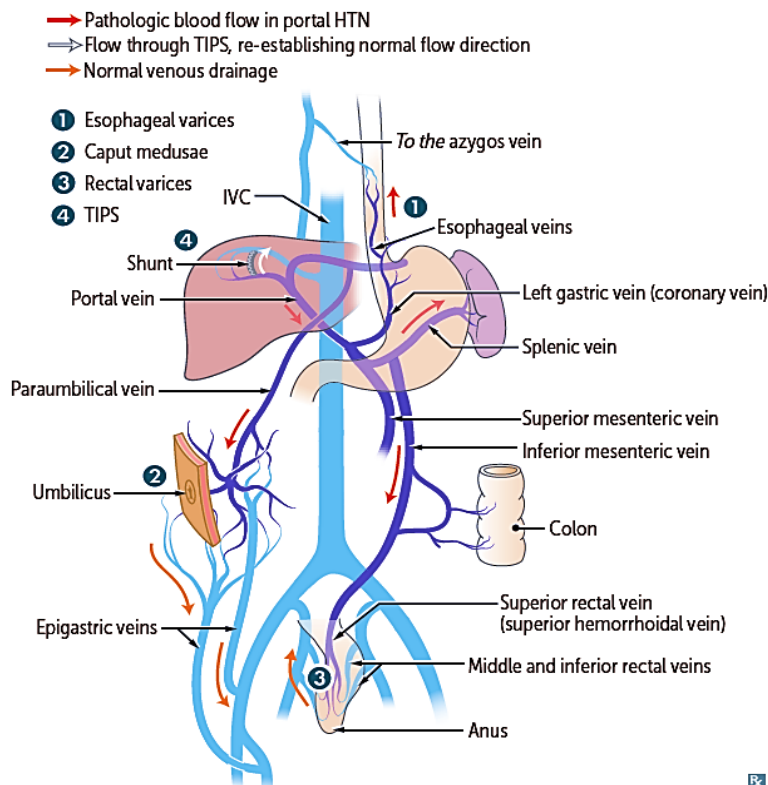


Figure 3: Portosystemic Anastomosis, Portal HTN

SITE OF ANASTOMOSIS	Clinical sign	Portal ↔ systemic
1 Esophagus	Esophageal varices	Left gastric ↔ esophageal
2 Umbilicus	Caput medusae	Paraumbilical ↔ small epigastric veins of the anterior abdominal wall
3 Rectum	Anorectal varices (not internal hemorrhoids)	Superior rectal ↔ middle and inferior rectal

Varices of the gut, butt and caput (medusae) are commonly seen with portal HTN (hypertension). Treatment with a trans-jugular intrahepatic portosystemic shunt (**TIPS**) between the portal vein

and hepatic vein relieves the portal HTN by shunting blood to systemic circulation, bypassing the liver – see 4 on the above figure (Figure 3).

Histology

Functional Histology of GIT and accessory glands of GIT

Digestive system: oral to anal canal + extrinsic glands

- ✓ It consists of a **hollow tube** (highly **modified in the oral cavity**) of varying diameter, composed of 4 layers: **mucosa**, **submucosa**, **muscularis externa**, and **serosa** (or **adventitia** if extraperitoneal)

Oral region:

Mucosa- epithelia + connective tissue

- ✓ Mostly lined by a **stratified squamous epithelium**

Lips

- ✓ **External** (skin) region, the **vermillion zone**, and **the internal** (mucosal) region.
- ✓ The first two regions are covered by **stratified squamous keratinized epithelium**, whereas the internal region is lined by a wet stratified squamous **non-keratinized epithelium**.

Palate

Parts: anterior hard palate (bony shelf in its core) and a posterior soft palate (skeletal muscle in its core).

The entire nasal aspect of the (hard & soft) palate (with the exception of the **uvula**) is lined by **pseudostratified ciliated columnar epithelium** (respiratory epithelium) while:

- ✓ **The hard palate** is lined on its oral aspect by stratified squamous keratinized epithelium.
- ✓ **The soft palate** is lined on its oral aspect by stratified squamous nonkeratinized epithelium.

Teeth

Components: internal soft tissue, the **pulp**, and **three calcified tissues:** enamel and cementum, and dentin. As in bone, calcium hydroxyapatite is the mineral material in the calcified dental tissues.

Parts: Teeth have an enamel-covered **crown**, a cementum-covered **root**, and a **cervix (neck)**, the region where the two surface materials meet.

1. Enamel

- ✓ **is the most calcified tissue in the body**
- ✓ **made of enamelin** -fibrous keratin like protein laid by **ameloblasts** at crown, and
- ✓ Becomes calcified by the deposition of calcium-hydroxyapatite crystals.
- ✓ Highly susceptible for acids- **dental caries**
- ✓ consists of only **4% organic** material and **96% inorganic** material
- ✓ **acellular after tooth eruption and therefore cannot repair itself**

2. dentin

- ✓ Calcified (72%) matrix containing **type I collagen fibers**; **harder than bone**
- ✓ Laid by **odontoblasts**, which **persist and continue to elaborate dentin** for the life of the tooth unlike ameloblasts.

- ✓ **Vascular**, sensitive to stimuli like pain
- ✓ **Highly resistant**; even remains after enamel and odontoblast destruction

3. Cementum

- a. **Type I collagen**—containing calcified matrix, which is produced by **cementoblasts**.
- b. continuously elaborated even after tooth eruption,
- c. Unlike bone, cementum is **avascular and lacks nerves**.
- d. Like bone, cementum is perforated by **Sharpey's fibres**

4. dental pulp

- ✓ A gelatinous, richly **vascularized** connective tissue containing **odontoblasts in its peripheral layer** (closest to the dentin), **fibroblasts and mesenchymal cells**, and **types I (60%) and III collagen-40%**
- ✓ **Contains afferent nerve fibers**. All sensations from the pulp are interpreted as pain in the central nervous system

Dental-supporting structures

- **The periodontal ligaments** - composed of a **type I (80%) & III (20%) collagen**, **vascularized** and **innervated**; for **shock absorption, proprioception, fixation**.
- **The Gingivae (gums)**- covered by stratified squamous keratinized or para-keratinized epithelium
- **Alveolar bone**-having an inner layer and outer layer of compact bone in contact with spongy bone

Tongue

Parts: anterior two-thirds and a posterior one-third by the V-shaped sulcus terminalis, whose apex ends in the foramen cecum.

- Its **dorsal surface**: stratified squamous **para-keratinized to keratinized** epithelium;
- its **ventral surface**: stratified squamous **non-keratinized epithelium**.

Lingual papillae are located on the **dorsal surface** of the **anterior two-thirds of the tongue**.

- a. **Filiform papillae**: **lack taste buds**.
- b. **Fungiform papillae**: they contain occasional **taste buds**.
- c. **Foliate papillae**: their **taste buds degenerate** shortly after the second year of life.
- d. **Circumvallate papillae**: lie just anterior to the sulcus terminalis; **possess well developed taste buds**.
 - Contain **Glands von Ebner**- deliver their **purely serous secretion** into each papilla

Alimentary canal: basic histology

- ✓ **Mucosa (mucous membrane)**
- Epithelium- all except esophagus and lower 2/3rd anal canal, are simple columnar.
- Lamina Propria.
- Muscularis Mucosae (interna)
- ✓ **Submucosa: C.T: loose + dense irreg. colla. /fibroelastic CT**
 - Submucosal plexus- “Plexus of Meissner” – for secretion
 - Has glands only in **esophagus & duodenum** (**brunners glands**)
- ✓ **Muscularis externa: 2 or 3 layers of muscles**
- Myenteric plexus- “Plexus of Auerbach” for motility
- 3 layers in stomach

✓ **Serosa /adventitia**

- Tunica adventitia – loose CT & Tunica serosa – covered by mesothelium

Four distinct types of mucosal variations:

- **Protective mucosa:** stratified squamous epithelium. In the oral cavity, pharynx, esophagus, and anal canal.
- **Secretory mucosa:** contain cells for secretion of digestive enzymes. Found in the **stomach & SI**.
- **Absorptive mucosa:** contains two key structures, crypts and villi, and is responsible primarily for absorbing digested nutrients. Found along the entirety of the small intestine.
- **Absorptive/protective mucosa:** specializes in water absorption and mucous secretion. Found in the LI.

Esophagus- 25 cm

- Lined by a stratified squamous non-keratinized epithelium.
- The submucosa contains mucus-secreting esophageal glands
- The upper third of the muscularis externa composed only of **skeletal muscle**; the middle third is composed of a combination of **smooth** and **skeletal** muscle; and the lower third is composed only of **smooth muscle**.
- Two physiological sphincters (the pharyngoesophageal and the gastroesophageal) in the **muscularis externa** ensure that the bolus is transported in toward the stomach only by peristalsis.

Stomach:

Mucosa: Simple columnar epithelium of the gastric mucosa is composed of mucinogen-producing surface lining cells, but they are **not goblet cells**

Gastric submucosa:

- houses Meissner (submucosal) plexus with arterial and venous plexuses

Gastric muscularis externa:

- Composed of **three layers** of smooth muscle: an **incomplete inner oblique layer**, a thick middle circular layer that forms the pyloric sphincter, and an outer longitudinal layer.
- **Auerbach's (myenteric) plexuses** located between the middle circular and outer longitudinal smooth muscle layers and responsible for Motility (mixing of gastric contents and emptying of the stomach).

Gastric glands: are simple **branched tubular glands** in the lamina propria of the **cardia, Fundus, and pylorus**.

Cells of the fundic/body glands and their secretions

- Parietal (oxyntic) cells: HCL/ Intrinsic factor
- Chief (peptic, zymogenic) cells: pepsinogen/lipase/renin
- Mucous neck cells: mucous
- Diffuse neuroendocrine system (DNES) cells: peptides. E.g. histamine,

Cardiac and pyloric glands: different from fundic glands in that they are coiled tubular **mucus-secreting** glands and **lack chief cells**

Region	Major Characteristics	Mucosal Cell Types	Function of Surface Mucosal Cells Surface
Stomach (body and fundus)	Rugae: shallow pits; deep glands	• Mucus cells • Chief cells • Parietal cells • Enteroendocrine (EE) cells	-Secrete mucus; form protective layer against acid; tight junctions between these cells probably contribute to the acid barrier of the epithelium. -Secrete pepsinogen and lipase precursor – chief cells -Secrete HCl and intrinsic factor -Secrete a variety of peptide hormones- DNES
Pylorus	Deep pits; shallow, branched glands	• Mucous cells • Parietal cells • EE cells	Same as above Same as above High concentration of gastrin
Small intestine	Villi, plicae, and crypts	Columnar absorptive cells	Contain numerous microvilli that greatly increase the luminal surface area, for absorption
1.Duodenum	Brunner glands, which discharge alkaline secretion	• Goblet cells • Paneth cells • EE cells	-Secrete acid glycoproteins that protect mucosal linings -Contains granules that contain Lysozyme. May play a role in regulating intestinal flora -High concentration of cells that secrete cholecystokinin and secretin
2.Jejunum	Villi, well developed plica, Crypts	Same cell types as found in the duodenal epithelium	Same as above
3.Ileum	Aggregations of lymph nodules called Peyer's patches	M cells found over lymphatic nodules and Peyer's Patches	Endocytose and transport antigen from the lumen to lymphoid cells
Large intestine	Lacks villi, crypts	Mainly mucus-secreting and absorptive cells	Transports Na ⁺ (actively) and water (passively) out of lumen

Table: Summarized Histology of GIT; adapted from Kaplan Anatomy Textbook

Small intestine.

- The luminal surface possesses **plicae circulares (2-3x)**, **intestinal villi (10x)**, and **microvilli (20x fold)**, which collectively increase the luminal surface area by a factor of **400 to 600**
- Unlike rugae, plica circularis (valves of Kerckring), these folds don't **flatten out** during fullness.

Mucosa of the small intestine:

Goblet cells: are unicellular glands that produce mucinogen, **increase in number** from **the duodenum to the ileum**

Surface absorptive cells: have a **glycocalyx**, which overlies the microvilli and binds various enzymes, including disaccharidases and dipeptidases

DNES cells: produce and secrete gastrin, cholecystokinin, gastric inhibitory peptide, and several other hormones

Lamina propria:

- Occupies the cores of the villi and the interstices between the numerous **glands(Crypts) of Lieberkühn.**
- Consists of loose connective tissue with lymphoid cells, fibroblasts, mast cells, smooth muscle cells, nerve endings, and lymphoid nodules.
- Also contains lacteals (blind-ended lymphatic vessels) and capillary loops.

Paneth cells: located at the base of the crypts of Lieberkühn, secrete the **antibacterial enzyme lysozyme** stored in large, apical, membrane-bounded secretory granules. These cells also release other antibacterial agents, **defensins** and **tumor necrosis factor alpha** and display extensive RER (basally), a large supranuclear Golgi complex, and many mitochondria.

Regenerative cells: located in the basal half of the crypts of Lieberkühn, are thin, tall, columnar stem cells that divide to replace themselves and the other types of epithelial cell (stem cell, like that of stomach w/c found at isthmus)

M (microfold) cells: are highly specialized, have an unusual shape, and lie in the epithelium over lymphoid nodules and Peyer patches. They are derived from undifferentiated cells of the crypts of Lieberkühn.

External layer of the small intestine:

- Serosa covers all of the jejunum and ileum and part of the duodenum.
- Adventitia covers the remainder of the duodenum
- **NB: sphincters are from inner circular layer of muscularis externa**

Large intestine.

- It houses bacteria that produce **vitamin B12 and vitamin K**
- Produces abundant mucus, which lubricates its lining and facilitates the passage of feces
- Functions primarily in the absorption of **electrolytes, fluids, and gases**
- Dead bacteria and indigestible remnants of the ingested material are compacted into feces.

Unique features of LI

- **No villi and Paneth cells**
- **No permanent circular folds**
- Smooth muscle forms
 - Taeniae coli; Haustra
- **Epiploic appendages** on serosal lining

Anal canal:

The anal mucosa displays longitudinal folds called **anal columns**, which join each other to form anal valves.

The regions between adjacent valves are known as **anal sinuses**.

Epithelium of the anal canal:

- ✓ Simple columnar changing to simple cuboidal proximal to the anal valves.
- ✓ Stratified squamous non-keratinized distal to the anal valves.
- ✓ Changes to stratified squamous keratinized (epidermis) at the anus.

Lamina propria and submucosae: hemorrhoids veins

Covered externally by **adventitia**.

Appendix

The wall is thickened by large aggregates of **lymphoid nodules** in the mucosa and even in the submucosa

Mucosa of the appendix:

The epithelium: simple columnar and contains surface columnar cells and goblet cells.

The lamina propria: displays numerous lymphoid nodules (capped by M cells) and lymphoid cells. It **does not form villi** but possesses **shallow crypts of Lieberkühn** with some goblet cells, surface columnar cells, regenerative cells, **occasional Paneth cells**, and numerous DNES cells, especially deep in the crypts; Completely covered by **serosa**

Accessory glands of GIT

Salivary Glands

Can be major – 3 pairs (in capsule) or minor-numerous (10% of saliva)

The secretory end pieces present two types of secretory cells-serous and mucous -as well as the nonsecretory myoepithelial cells.

Serous cells:

- ✓ Exhibit characteristics of polarized **protein-secreting cells**; protective function.

Mucous cells:

- ✓ They exhibit the characteristics of **mucus-secreting cells**, containing glycoproteins important for the moistening and lubricating functions of the saliva.
- ✓ Most of these glycoproteins are called mucins and contain 70-80% carbohydrate moieties in their structure.

Myoepithelial cells:

- ✓ surround each secretory portion, usually two to three cells per secretory unit, are well developed and branched (and are sometimes called **basket cells**), whereas those associated with intercalated ducts are **spindle shaped**
- ✓ function: squeezing by **contraction** and mainly for **prevention of end piece distention**

Secretion path: **secretory end piece → intercalated duct → striated duct → interlobular (excretory) duct**

- ✓ **Striated duct possesses ion transporting cells**

Regulation:

- ✓ *Parasympathetic stimulation, usually through the smell or taste of food, promotes vasodilation and a **copious watery secretion**.*
- ✓ *Sympathetic stimulation produces **small amounts of viscous saliva**, rich in organic material.*
- ✓ **Saliva is hypotonic normally.**

Parotid Gland- 25%

- ✓ Branched acinar gland; exclusively of serous cells containing secretory granules that are rich in proteins and have a high amylase activity.
- ✓ This activity is responsible for most of the hydrolysis of ingested carbohydrates by **ptyalin**.
- ✓ As in other large salivary glands, the CT contains many plasma cells and lymphocytes. The plasma cells secrete **IgA**

Submandibular (Submaxillary) Gland- 70%

- ✓ branched tubuloacinar gland; mixed: contains both mucous and serous cells.
- ✓ 90% of the end pieces are serous acinar, whereas 10% consist of mucous tubules with serous demilunes.
- ✓ The presence of extensive lateral and basal membrane infoldings toward the vascular bed increases the ion-transporting surface area 60 times, facilitating electrolyte and water transport.
- ✓ Serous cells are responsible for the weak **amylolytic** activity present in this gland and its saliva.
- ✓ The cells that form the **demilunes** secrete the enzyme **lysozyme**, whose main activity is to hydrolyze the walls of certain bacteria. Some acinar and intercalated duct cells secrete **lactoferrin**

Sublingual Gland- 5%

- Like the submandibular gland, is a branched tubuloacinar gland formed of serous and mucous cells.
- Mucous cells predominate in this gland; serous cells are present almost exclusively on demilunes of mucous tubules.
- As in the submandibular gland, cells that form the demilunes in this gland secrete lysozyme.
- Intralobular ducts are not as well developed as in other major salivary glands.

Minor Salivary Glands

- ✓ Although variations exist, minor salivary glands are usually mucous. Non-encapsulated.
- ✓ The small serous glands present in the posterior region of the tongue (**von Ebner's glands**) are the only exception.
- ✓ Lymphocyte aggregates are commonly observed within minor salivary glands, associated with IgA secretion.

Pancreas:

Both Exocrine and endocrine

Exocrine portion: serous compound tubuloacinar gland

- Zymogen (secretory) granules are membrane bound and densely packed in the apical region of pancreatic acinar cells.
- Their basal plasmalemma has receptors for cholecystokinin and acetylcholine

Pancreatic duct:

- The initial intra-acinar portion of the intercalated ducts is formed by centroacinar cells, which are low cuboidal with a pale cytoplasm.
- From the initial portion, the intercalated ducts converge into a small number of intralobular ducts, which in turn empty into large interlobular ducts that empty into the main (or accessory) pancreatic duct

Pancreatic exocrine secretion:

a. Enzyme-poor alkaline fluid

- Enzyme-poor alkaline fluid is released in large quantities by intercalated duct cells
- Stimulated by secretin in conjunction with acetylcholine.
- Function. It probably neutralizes the acidic chyme as it enters the duodenum.

b. Digestive enzymes

- Digestive enzymes are synthesized and stored in the pancreatic acinar cells.
- Their release is stimulated by cholecystokinin, previously known as pancreozymin and costimulated by acetylcholine released by postganglionic parasympathetic fibers.
- Enzymes include pancreatic amylase, pancreatic lipases, ribonuclease, and deoxyribonuclease; proenzymes include trypsinogen, chymotrypsinogen, procarboxypeptidase, and proelastase

Islets of Langerhans (endocrine pancreas):

They produce several polypeptide hormones, but each cell type produces only one hormone. Islet hormones:

- Glucagon is produced by **alpha-cells**, at **periphery**. 20%,
- Insulin is produced by **beta-cells** and **central**. 70%.
- Somatostatin is produced by **delta-cells**. It inhibits release of hormones by nearby secretory cells and reduces motility of the gastrointestinal tract and gallbladder, 5%
- Gastrin, produced by **G cells**, stimulates (in conjunction with histamine and Ach) gastric HCl secretion.
- Pancreatic polypeptide, produced by **PP cells**, inhibits release of exocrine pancreatic secretions.

Liver

- Composed of a single type of parenchymal cell, the hepatocyte.
- It is surrounded by a dense, irregular collagenous connective tissue known as **Glisson capsule**, which gives rise to septa that subdivide the liver into lobes and lobules
- **Liver lobules:**

i. The classic liver lobule: a hexagonal mass of tissue primarily composed of plates of hepatocytes, which radiate from the region of the central vein toward the periphery

ii. Portal areas - (portal canals or **portal triads**)

1. The portal vein,
2. Hepatic artery,
3. Lymph vessel, and bile duct.

iii. Liver sinusoids-

- **Fenestrated** capillaries that arise at the periphery of a lobule and run between adjacent plates of hepatocytes. They receive blood from the vessels in the portal areas and deliver it to the central vein. They lack basal laminae.
- contain phagocytic cells (**Kupffer cells**) derived from monocytes; these cells remove debris, old erythrocytes, and cellular fragments from the bloodstream

iv. Space of Disse

- sub endothelial space between hepatocytes and sinusoidal lining cells.
- Contains the short microvilli of hepatocytes, and occasional nonmyelinated nerve fibers.
- also contains **stellate fat-storing cells (Ito cells, perisinusoidal stellate cells)**, which preferentially store **vitamin A**.
- When the liver is compromised, hepatocytes release **tumor growth factor beta** and, in response, these **fat-storing cells** can divide, **change their phenotype**, and begin to synthesize **collagen, leading to fibrosis** and, **if necessary, differentiate into myofibroblasts** to control blood flow into the sinusoids.

v. Portal lobule:

- a. A triangular region with three apices that are neighboring central veins and a center in a portal area. It contains portions of three adjacent classic liver lobules.
- b. The portal lobule is **defined in terms of bile flow**. The bile duct is in the center of this lobule.

vi. Hepatic acinus (of Rappaport):

- A diamond-shaped region encompassing triangular sections of two adjacent classic liver lobules (with apices that are the central veins) and is divided by the common distributing vessels.
- This concept of liver lobulation is **defined in terms of blood flow** from the distributing vessels in a single portal area, so, determines the pathology of blood.
- Can be divided into three zones (1, 2 & 3) on the basis of the proximity of the hepatocytes to the **incoming blood**. eg. *zone 1-periphery (more blood supply) ...*

Blood and bile flow:

Blood flow into the liver is derived from two sources toward the central vein. The hepatic artery: oxygen-rich blood from the abdominal aorta and supplies 20% to 30%. The portal vein: nutrient-rich blood from the GIT and spleen; it supplies 70% to 80%. Blood flow out of the liver occurs via the hepatic vein, formed by the union of numerous sublobular veins, which collect blood from the central veins.

Bile flow is directed toward the periphery of the classic liver lobule (**in the opposite direction of blood flow**). Bile is carried in a system of ducts that culminate in the left and right hepatic ducts, which leave the liver and carry bile to the gallbladder.

Bile ducts:

They receive bile from the **canals of Hering** (bile ductules of liver). They enlarge and fuse to form the hepatic ducts, which leave the liver at the porta hepatis.

Hepatocyte:

One round central nucleus; about 25% of the cells are bi-nucleated. Occasionally, nuclei are polyploid, surface with microvilli toward canaliculi

Functions:

- **Exocrine secretion**- involves the production and release of 600 to 1200 mL of bile per day.
- **Endocrine secretion**- involves the production and release of several plasma proteins (e.g., prothrombin, fibrinogen, albumin, factor III, and lipoproteins) and urea.
- **Metabolite storage**- in the form of glycogen (stored glucose) and triglycerides (stored lipid).
- **Gluconeogenesis**- the conversion of amino acids and lipids into glucose, a complex process catalyzed by a series of enzymes.
- **Detoxification** entails the inactivation of various substances, such as drugs, noxious chemicals, and toxins, by enzymes, such as the microsomal mixed-function oxidase system, that catalyze the oxidation, methylation, or conjugation of such substances
- **IgA transfer**- involves the uptake of IgA across the space of Disse and its release into bile canaliculi. IgA is transported through the hepatobiliary duct system to the intestine

Gallbladder:

- The mucosa is composed of a simple columnar epithelium and a richly vascularized lamina propria
- The muscle layer is composed of a thin, oblique layer of smooth muscle cells.
- The connective tissue layer consists of dense irregular collagenous connective tissue and houses nerves and blood vessels.
- The serosa covers most of the gallbladder, but adventitia is present where the organ is attached to the liver
- NB: **muscularis layer possess receptors of CCK & Ach for contraction.**

Physiology

Regulation of the GI System

Nervous Regulation

Extrinsic (sympathetic and parasympathetic) innervation of the GIT has been discussed in Anatomy. Here, the enteric nervous system (the intrinsic innervation) will be discussed.

Intrinsic innervation (enteric nervous system)

- coordinates and relays information from the parasympathetic and sympathetic nervous systems to the GI tract.
- uses **local reflexes** to relay information within the GI tract.
- controls most functions of the GI tract, especially motility and secretion, even in the absence of extrinsic innervation.
- a. **Myenteric plexus (Auerbach's plexus)**
 - primarily controls the motility of the GI smooth muscle.
- b. **Submucosal plexus (Meissner's plexus)**
 - primarily controls secretion and blood flow.
 - receives sensory information from chemoreceptors and mechanoreceptors in the GI tract.

Hormonal Regulation of GIT

A. GI Hormones

- are released from endocrine cells in the GI mucosa into the portal circulation, enter the general circulation, and have physiologic actions on the target cells.
- Four substances meet the requirements to be considered "official" GI hormones; others are considered "candidate" hormones. The four official GI hormones are gastrin, cholecystokinin (CCK), secretin, and glucose-dependent insulintropic peptide (GIP).

SUMMARY OF GI HORMONES			
Hormones	Site of Secretion	Stimulus for Secretion	Action
Gastrin	G cells of stomach	Small peptides & amino acids; Distension of stomach; Vagus (via GRP). Inhibited by H ⁺ in stomach Inhibited by somatostatin	↑ Gastric H ⁺ secretion Stimulates growth of gastric mucosa
CCK	I cells of duodenum and jejunum	Small peptides & amino acids Fatty acids	Stimulates contraction of gallbladder & relaxation of sphincter of Oddi ↑ Pancreatic enzyme and HCO ₃ ⁻ secretion ↑ Growth exocrine pancreas/gallbladder Inhibits gastric emptying
Secretin	S cells of duodenum	H ⁺ in duodenum Fatty acids in duodenum	↑ Pancreatic enzyme and HCO ₃ ⁻ secretion Biliary HCO ₃ ⁻ secretion ↓ Gastric H ⁺ secretion
GIP	Duodenum & jejunum	Fatty acids, amino acids and oral glucose	↑ Insulin secretion ↓ Gastric H ⁺ secretion

CCK = cholecystokinin; GIP = glucose-dependent insulinotropic peptide; GRP = gastrin-releasing peptide.

Gastrointestinal Motility

- Contractile tissue of the GI tract is almost exclusively **unitary smooth muscle**, with the exception of the pharynx, upper one-third of the esophagus, and external anal sphincter, all of which are **striated muscle**.
- Depolarization of **circular muscle** leads to contraction of a ring of smooth muscle and a **decrease in diameter** of that segment of the GI tract.
- Depolarization of **longitudinal muscle** leads to contraction in the longitudinal direction and a **decrease in length** of that segment of the GI tract.
- Phasic contractions** occur in the esophagus, gastric antrum, and small intestine, which contract and relax periodically.
- Tonic contractions** occur in the lower esophageal sphincter, and ileocecal and internal anal sphincters.

A. Slow waves

- are **oscillating membrane potentials** inherent to the smooth muscle cells of some parts of the GI tract.
- occur spontaneously.
- originate in the **interstitial cells of Cajal**, which serve as the **pacemaker** for GI smooth muscle.
- are **not action potentials**, although they **determine the pattern of action potentials** and, therefore, the pattern of contraction.

1. Mechanism of slow wave production

- is the cyclic opening of Ca²⁺ channels (depolarization) followed by opening of K⁺ channels (repolarization).
- Depolarization during each slow wave** brings the membrane potential of smooth muscle cells closer to threshold and, therefore, **increases the probability that action potentials will occur**.
- Action potentials, produced on top of the background of slow waves, then initiate phasic contractions of the smooth muscle cells.

2. Frequency of slow waves

- varies along the GI tract, but is constant and characteristic for each part of the GI tract.
- is *not* influenced by neural or hormonal input. In contrast, the frequency of the action potentials that occur on top of the slow waves is modified by neural and hormonal influences.
- **sets the maximum frequency of contractions** for each part of the GI tract.
- is **lowest in the stomach** (3 slow waves/min) and **highest in the duodenum** (12 slow waves/min).

B. Chewing, swallowing, and esophageal peristalsis

1. Chewing

- lubricates food by mixing it with saliva.
- decreases the size of food particles to facilitate swallowing and to begin the digestive process.

2. Swallowing

- The swallowing reflex is **coordinated in the medulla**. Fibers in the vagus and glossopharyngeal nerves carry information between the GI tract and the medulla.
- The following sequence of events is involved in swallowing:
 - a. The nasopharynx closes and, at the same time, **breathing is inhibited**.
 - b. The laryngeal muscles contract to close the glottis and elevate the larynx.
 - c. **Peristalsis begins in the pharynx** to propel the food bolus toward the esophagus. Simultaneously, the **upper esophageal sphincter relaxes** to permit the food bolus to enter the esophagus.

3. Esophageal motility

- The esophagus propels the swallowed food into the stomach.
- Sphincters at either end of the esophagus prevent air from entering the upper esophagus and gastric acid from entering the lower esophagus.
- Because the esophagus is located in the thorax, intraesophageal pressure equals thoracic pressure, which is **lower than atmospheric pressure**. In fact, a balloon catheter placed in the esophagus can be used to measure intrathoracic pressure.
- The following sequence of events occurs as food moves into and down the esophagus:
 - a. As part of the swallowing reflex, the **upper esophageal sphincter** relaxes to permit swallowed food to enter the esophagus.
 - b. The upper esophageal sphincter then contracts so that food will not reflux into the pharynx.
 - c. A **primary peristaltic contraction** creates an area of high pressure behind the food bolus. The peristaltic contraction moves down the esophagus and propels the food bolus along. **Gravity** accelerates the movement.
 - d. A **secondary peristaltic contraction** clears the esophagus of any remaining food.
 - e. As the food bolus approaches the lower end of the esophagus, the **lower esophageal sphincter relaxes**. This relaxation is vagally mediated, and the neurotransmitter is **VIP**.
 - f. The orad region of the stomach relaxes ("**receptive relaxation**") to allow the food bolus to enter the stomach.

4. Clinical correlations of esophageal motility

- a. **Gastroesophageal reflux (heartburn)** may occur if the tone of the lower esophageal sphincter is decreased and gastric contents reflux into the esophagus.
- b. **Achalasia** may occur if the lower esophageal sphincter does not relax during swallowing and food accumulates in the esophagus.

C. Gastric motility

- The stomach has three layers of smooth muscle—the usual longitudinal and circular layers, and a third oblique layer.
- The stomach has three anatomic divisions—the fundus, body, and antrum.
- The orad region of the stomach includes the fundus and the proximal body. This region contains oxyntic glands and is responsible for receiving the ingested meal.
- The caudad region of the stomach includes the antrum and the distal body. This region is responsible for the contractions that mix food and propel it into the duodenum.

1. "Receptive relaxation"

- is a vagovagal reflex that is initiated by distention of the stomach and is abolished by vagotomy.
- The oral region of the stomach relaxes to accommodate the ingested meal.
- CCK participates in “receptive relaxation” by increasing the distensibility of the oral stomach.

2. Mixing and digestion

- The caudad region of the stomach contracts to mix the food with gastric secretions and begins the process of digestion. The size of food particles is reduced.
- a. **Slow waves** in the caudad stomach occur at a frequency of 3–5 waves/min. They depolarize the smooth muscle cells.
- b. If threshold is reached during the slow waves, action potentials are fired, followed by contraction. Thus, the frequency of slow waves sets the maximal frequency of contraction.
- c. A **wave of contraction** closes the distal antrum. Thus, as the caudad stomach contracts, food is propelled back into the stomach to be mixed (**retropulsion**).
- d. Gastric contractions are **increased by vagal stimulation and decreased by sympathetic stimulation**.
- e. Even during fasting, contractions (the “**migrating myoelectric complex**”) occur at 90-minute intervals and clear the stomach of residual food. Motilin is the mediator of these contractions.

3. Gastric emptying

- The caudad region of the stomach contracts to propel food into the duodenum.
- a. The rate of **gastric emptying is fastest** when the stomach contents are **isotonic**. If the stomach contents are hypertonic or hypotonic, gastric emptying is slowed.
- b. **Fat inhibits gastric emptying** (i.e., increases gastric emptying time) by stimulating the release of **CCK**.
- c. **H⁺ in the duodenum inhibits gastric emptying** via direct neural reflexes. H⁺ receptors in the duodenum relay information to the gastric smooth muscle via interneurons in the GI plexuses.

D. Small intestinal motility

- The small intestine functions in the **digestion and absorption** of nutrients. The small intestine mixes nutrients with digestive enzymes, exposes the digested nutrients to the absorptive mucosa, and then propels any nonabsorbed material to the large intestine.
- As in the stomach, **slow waves** set the basic electrical rhythm, which occurs at a frequency of 12 waves/min. Action potentials occur on top of the slow waves and lead to contractions.
- Parasympathetic stimulation increases intestinal smooth muscle contraction; sympathetic stimulation decreases it.

1. Segmentation contractions

- **mix the intestinal contents.**
- A section of small intestine contracts, sending the intestinal contents (chyme) in both oral and caudad directions. That section of small intestine then relaxes, and the contents move back into the segment.
- This **back-and-forth movement** produced by segmentation contractions causes mixing without any net forward movement of the chyme.

2. Peristaltic contractions

- are highly coordinated and **propel the chyme** through the small intestine toward the large intestine. Ideally, peristalsis occurs after digestion and absorption have taken place.
- **Contraction behind the bolus** and, simultaneously, **relaxation in front of the bolus** cause the chyme to be propelled caudally.
- The peristaltic reflex is **coordinated by the enteric nervous system**.

3. Gastroileal reflex

- is mediated by the extrinsic ANS and possibly by gastrin.
- The presence of food in the stomach triggers increased peristalsis in the ileum and relaxation of the ileocecal sphincter. As a result, the intestinal contents are delivered to the large intestine.

E. Large intestinal motility

- Fecal material moves from the cecum to the colon (i.e., through the ascending, transverse, descending, and sigmoid colons), to rectum, and then to the anal canal.
- **Haustra**, or saclike segments, appear after contractions of the large intestine.

1. Cecum and proximal colon

- When the proximal colon is distended with fecal material, the ileocecal sphincter contracts to prevent reflux into the ileum.
 - a. **Segmentation contractions** in the proximal colon mix the contents and are responsible for the appearance of haustra.
 - b. **Mass movements occur 1 to 3 times/day** and cause the colonic contents to move distally for long distances (e.g., from the transverse colon to the sigmoid colon).

2. Distal colon

- Because most colonic water absorption occurs in the proximal colon, fecal material in the distal colon becomes semisolid and moves slowly. Mass movements propel it into the rectum.

3. Rectum, anal canal, and defecation

- The sequence of events for defecation is as follows:
 - a. As the rectum fills with fecal material, it contracts and the internal anal sphincter relaxes (rectosphincteric reflex).
 - b. Once the rectum is filled to about 25% of its capacity, there is an urge to defecate. However, defecation is prevented because the external anal sphincter is tonically contracted.
 - c. **When it is convenient to defecate**, the external anal sphincter is relaxed voluntarily. The smooth muscle of the rectum contracts, forcing the feces out of the body.
 - Intra-abdominal pressure is increased by expiring against a closed glottis (**Valsalva maneuver**).

4. Gastrocolic reflex

- The presence of **food in the stomach** increases the motility of the colon and **increases the frequency of mass movements**.
 - a. The gastrocolic reflex has a rapid **parasympathetic** component that is initiated when the stomach is stretched by food.
 - b. A slower, hormonal component is mediated by CCK and gastrin.

5. Disorders of large intestinal motility

- a. Emotional factors strongly influence large intestinal motility via the extrinsic ANS. **Irritable bowel syndrome** may occur during periods of stress and may result in **constipation** (increased segmentation contractions) or **diarrhea** (decreased segmentation contractions).
- b. Megacolon (Hirschsprung's disease), the absence of the colonic enteric nervous system, results in constriction of the involved segment, marked dilatation and accumulation of intestinal contents proximal to the constriction, and severe constipation.

F. Vomiting

- A wave of reverse peristalsis begins in the small intestine, moving the GI contents in the oral direction.
- The gastric contents are eventually pushed into the esophagus. If the upper esophageal sphincter remains closed, **retching** occurs. If the pressure in the esophagus becomes high enough to open the upper esophageal sphincter, **vomiting** occurs.
- The **vomiting center** in the **medulla** is stimulated by tickling the back of the throat, gastric distention, and vestibular stimulation (motion sickness).

- The **chemoreceptor trigger zone** in the fourth ventricle is activated by emetics, radiation, and vestibular stimulation.

Secretion

A. Salivary secretion

1. Functions of saliva

- a. **Initial starch digestion** by α -amylase (ptyalin) and **initial triglyceride digestion** by lingual lipase.
- b. **Lubrication** of ingested food by mucus
- c. **Protection** of the mouth and esophagus by dilution and buffering of ingested foods

2. Composition of saliva

- a. Saliva is characterized by:
 - (1) High volume (relative to the small size of the salivary glands)
 - (2) *High K^+ and HCO_3^- concentrations*
 - (3) Low Na^+ and Cl^- concentrations
 - (4) **Hypotonicity**
 - (5) Presence of α -amylase, lingual lipase, and kallikrein
- b. The composition of saliva varies with the salivary flow rate.
 - (1) *At the lowest flow rates*, saliva has the lowest osmolarity and lowest Na^+ , Cl^- , and HCO_3^- concentrations, but has the highest K^+ concentration.
 - (2) *At the highest flow rates* (up to 4 mL/min), the composition of saliva is closest to that of plasma.

3. Formation of saliva

- Saliva is formed by three major glands—the **parotid, submandibular, & sublingual glands**.
 - The **structure** of each gland is similar to a bunch of grapes. The **acinus** (the blind end of each duct) is lined with acinar cells and secretes an initial saliva. A **branching duct system** is lined with columnar epithelial cells, which modify the initial saliva.
 - When saliva production is stimulated, **myoepithelial cells**, which line the acinus and initial ducts, contract and eject saliva into the mouth.
- a. The acinus
 - **produces an initial saliva** with a composition **similar to plasma**.
 - This initial saliva is **isotonic** and has the same Na^+ , K^+ , Cl^- , and HCO_3^- concentrations as plasma.
 - b. The ducts – modify the initial saliva and make it hypotonic. (because the ducts are relatively impermeable to water, they reabsorb more solute from the saliva than water).

4. Regulation of saliva production

- Saliva production is unique in that it is increased by **both the sympathetic and parasympathetic nervous system** (CN VII & IX). Parasympathetic activity is more important, however. (GI hormones have no role in salivary secretion).

B. Gastric Secretion

1. Gastric cell types and their secretions

- **Parietal cells**, located in the body, secrete **HCl** and **intrinsic factor**.
- **Chief cells**, located in the body, secrete **pepsinogen**.
 - **G cells**, located in the antrum, secrete gastrin.

2. Three phases of gastric secretion

- a. **Cephalic phase** –the sight, smell, taste, or thought of food stimulates secretion.
- b. **Gastric phase** –secretion is caused by the entry of food into the stomach.
- c. **Intestinal phase** –food entering the intestine causes feedback stimulation of gastric secretion.

2. Mechanism of gastric H^+ secretion

- Parietal cells **secrete HCl into the lumen of the stomach** and, concurrently, **absorb HCO_3^-** into the bloodstream as follows:

- a. In the parietal cells, CO_2 and H_2O are converted to H^+ and HCO_3^- , catalyzed by **carbonic anhydrase**.
- b. **H^+ is secreted into the lumen of the stomach** by the H^+-K^+ pump (**H^+ , K^+ -ATPase**). Cl^- is secreted along with H^+ ; thus, the secretion product of the parietal cells is HCl .
- c. The drug **omeprazole** (a “proton pump inhibitor”) inhibits the H^+ , K^+ -ATPase and blocks H^+ secretion.
- d. The **HCO_3^-** produced in the cells is absorbed into the bloodstream in exchange for Cl^- (Cl^- – HCO_3^- exchange). As HCO_3^- is added to the venous blood, the pH of the blood increases (“alkaline tide”). (Eventually, this HCO_3^- will be secreted in pancreatic secretions to neutralize H^+ in the small intestine.)
 - If vomiting occurs, gastric H^+ never arrives in the small intestine, there is no stimulus for pancreatic HCO_3^- secretion, and the arterial blood becomes alkaline (**metabolic alkalosis**).

3. Stimulation of gastric H^+ secretion

a. Vagal stimulation

- increases H^+ secretion by a direct pathway and an indirect pathway.
- In the direct path, the **vagus nerve innervates parietal cells** and stimulates H^+ secretion directly. The neurotransmitter at these synapses is **ACh**, the receptor on the parietal cells is muscarinic (M_3), and the second messengers for CCK are IP_3 and increased intracellular $[\text{Ca}^{2+}]$.
- In the indirect path, the vagus nerve innervates G cells and stimulates gastrin secretion, which then stimulates H^+ secretion by an endocrine action. The neurotransmitter at these synapses is **GRP** (not ACh).
- **Atropine**, a cholinergic muscarinic antagonist, inhibits H^+ secretion by blocking the direct pathway, which uses ACh as a neurotransmitter. However, atropine does not block H^+ secretion completely because it does not inhibit the indirect pathway, which uses **GRP** as a neurotransmitter.
- **Vagotomy** eliminates both direct and indirect pathways.

b. Gastrin

- is released in response to eating a meal (small peptides, distention of the stomach, vagal stimulation).
- stimulates H^+ secretion by interacting with the cholecystokinin_B (CCK_B) receptor on the parietal cells.
- The second messenger for gastrin on the parietal cell is $\text{IP}_3/\text{Ca}^{2+}$.

c. Histamine

- is released from enterochromaffin-like (ECL) cells in the gastric mucosa and diffuses to the nearby parietal cells.
- stimulates H^+ secretion by activating **H_2 receptors** on the parietal cell membrane.
- The H_2 receptor is coupled to adenylyl cyclase via a G_s protein.
- The second messenger for histamine is **cAMP**.
- H_2 receptor–blocking drugs, such as **cimetidine**, inhibit H^+ secretion by blocking the stimulatory effect of histamine.

d. Potentiating effects of ACh, histamine, and gastrin on H^+ secretion

- **Potentiation** occurs when the response to simultaneous administration of two stimulants
- is greater than the sum of responses to either agent given alone. As a result, low concentrations of stimulants given together can produce maximal effects.
- Potentiation of gastric H^+ secretion can be explained, in part, because **each agent has a different mechanism** of action on the parietal cell.

(1) Histamine potentiates the actions of ACh and gastrin in stimulating H^+ secretion.

- Thus, H_2 receptor blockers (e.g., **cimetidine**) are particularly effective in treating ulcers because they block both the direct action of histamine on parietal cells and the potentiating effects of histamine on ACh and gastrin.

(2) ACh potentiates the actions of histamine and gastrin in stimulating H^+ secretion.

- Thus, muscarinic receptor blockers, such as **atropine**, block both the direct action of ACh on parietal cells and the potentiating effects of ACh on histamine and gastrin.

4. Inhibition of gastric H⁺ secretion

- **Negative feedback** mechanisms inhibit the secretion of H⁺ by the parietal cells.
 - a. **Low pH (<3.0) in the stomach**
 - **Inhibits gastrin secretion** (negative feedback) and thereby inhibits H⁺ secretion
 - b. **Somatostatin**
 - Inhibits gastric H⁺ secretion directly as well as indirectly.
 - Direct pathway (somatostatin binds to receptors on parietal cells which are G_i coupled and decreases cAMP & hence antagonizes stimulatory action of histamine)
 - Indirect Pathway (by inhibiting release of gastrin & histamine).
 - c. **Prostaglandins**
 - **Inhibit** gastric H⁺ secretion by activating G_i, inhibiting adenylyl cyclase and **decreasing cAMP levels**.

5. Peptic ulcer disease

- is an ulcerative lesion of the gastric or duodenal mucosa.
- can occur when there is **loss of the protective mucous barrier** (of mucus and HCO₃⁻) and/or **excessive secretion of H⁺ and pepsin**.
- **Protective factors** are mucus, HCO₃⁻, prostaglandins, mucosal blood flow, and growth factors.
- **Damaging factors** are H⁺, pepsin, *Helicobacter pylori* (*H. pylori*), non-steroidal anti-inflammatory drugs (NSAIDs), stress, smoking and alcohol.

 - a. **Gastric ulcers**
 - The **gastric** mucosa is damaged.
 - Gastric **H⁺ secretion is decreased** because secreted H⁺ leaks back through the damaged gastric mucosa.
 - **Gastrin levels are increased** because decreased H⁺ secretion stimulates gastrin secretion.
 - A major cause of gastric ulcer is the gram-negative bacterium ***H. Pylori***.
 - *H. Pylori* colonizes the gastric mucus and releases cytotoxins that damage gastric mucosa.
 - *H. Pylori* contains urease, which converts urea to NH₃, thus alkalinizing the local environment and permitting *H. Pylori* to survive in the otherwise acidic gastric lumen.
 - b. **Duodenal ulcers**
 - The **duodenal** mucosa is damaged.
 - Gastric **H⁺ secretion is increased**. Excess H⁺ is delivered to duodenum, damaging the duodenal mucosa.
 - **Gastrin secretion in response to a meal is increased** (although baseline gastrin may be normal).
 - *H. Pylori* is also a major cause of duodenal ulcer. *H. pylori* inhibits somatostatin secretion (thus stimulating gastric H⁺ secretion) and inhibits intestinal HCO₃⁻ secretion (so there is insufficient HCO₃⁻ to neutralize the H⁺ load from the stomach).
 - c. **Zollinger–Ellison syndrome**
 - occurs when a gastrin-secreting tumor of the pancreas causes increased H⁺ secretion.
 - H⁺ secretion continues unabated because the gastrin secreted by pancreatic tumor cells is not subject to negative feedback inhibition by H⁺.

6. Drugs that block gastric H⁺ secretion

- a. **Atropine**
 - Blocks H⁺ secretion by inhibiting cholinergic muscarinic receptors on parietal cells, thereby inhibiting ACh stimulation of H⁺ secretion.
- b. **Cimetidine**
 - Blocks H₂ receptors and thereby inhibits histamine stimulation of H⁺ secretion.

- Is particularly effective in reducing H^+ secretion because it not only blocks the histamine stimulation of H^+ secretion, but also blocks histamine's potentiation of ACh effects.

c. Omeprazole

- Is a proton pump inhibitor.
- Directly inhibits H^+ , K^+ -ATPase and H^+ secretion.

C. Pancreatic secretion

- contains a high concentration of HCO_3^- , whose purpose is to neutralize the acidic chyme that reaches the duodenum.
- contains enzymes essential for the digestion of protein, carbohydrate, and fat.

1. Composition of pancreatic secretion

a. Pancreatic juice is characterized by:

- (1) High volume
- (2) Virtually the same Na^+ and K^+ concentrations as plasma
- (3) Much higher HCO_3^- concentration than plasma
- (4) Much lower Cl^- concentration than plasma
- (5) **Isotonicity**
- (6) Pancreatic lipase, amylase, and proteases

b. The composition of the aqueous component of pancreatic secretion varies with the flow rate

- At **low flow rates**, the pancreas secretes an isotonic fluid that is composed mainly of Na^+ and Cl^- .
- At **high flow rates**, the pancreas secretes an isotonic fluid that is composed mainly of Na^+ and HCO_3^- .
- Regardless of the flow rate, pancreatic secretions are **isotonic**.

2. Formation of pancreatic secretion

- Like the salivary glands, the exocrine pancreas resembles a bunch of grapes.
- The acinar cells of the exocrine pancreas make up most of its weight.

a. Acinar cells

- produce a small volume of initial pancreatic secretion, which is mainly Na^+ and Cl^- .

b. Ductal cells

- modify the initial pancreatic secretion by **secreting HCO_3^-** and **absorbing Cl^-** via a Cl^- – HCO_3^- **exchange** mechanism in the luminal membrane.
- Because the pancreatic ducts are **permeable to water**, H_2O moves into the lumen to make the pancreatic secretion isosmotic.

3. Stimulation of pancreatic secretion

a. Secretin

- is secreted by the S cells of the duodenum in response to H^+ in the duodenal lumen
- acts on the pancreatic **ductal cells** to increase **HCO_3^- secretion**.
- Thus, when H^+ is delivered from the stomach to the duodenum, secretin is released. As a result, HCO_3^- is secreted from the pancreas into the duodenal lumen to neutralize the H^+ .
- The second messenger for secretin is **cAMP**.

b. CCK

- is secreted by the I cells of the duodenum in response to small peptides, amino acids, and fatty acids in the duodenal lumen.
- Acts on the pancreatic **acinar cells** to increase enzyme secretion (amylase, lipases, proteases).
- Potentiates the effect of secretin on the ductal cells to stimulate HCO_3^- secretion.
- The second messengers for CCK are IP_3 and increased intracellular $[Ca^{2+}]$.

c. ACh

- Stimulates enzyme secretion by the acinar cells and, like CCK, potentiates the effect of secretin on HCO_3^- secretion.

d. Cystic fibrosis

- Is a disorder of pancreatic secretion.
- Results from a defect in Cl^- channels that is caused by a mutation in **cystic fibrosis transmembrane conductance regulator gene (CFTR) gene**.
- Is associated with a **deficiency of pancreatic enzymes** resulting in malabsorption and steatorrhea.

D. Bile secretion and gallbladder function

1. Composition and function of bile

- Bile contains **bile salts**, phospholipids, cholesterol, and bile pigments (bilirubin).

a. Bile salts

- are **amphipathic** molecules because they have both hydrophilic and hydrophobic portions. In aqueous solution, bile salts orient themselves around droplets of lipid and keep the lipid droplets dispersed (**emulsification**).
- aid in the intestinal digestion and absorption of lipids by emulsifying and solubilizing them in **micelles**.

b. Micelles

- Above a **critical micellar concentration**, bile salts form micelles.
- Bile salts are positioned on the outside of the micelle, with their hydrophilic portions dissolved in the aqueous solution of the intestinal lumen and their hydrophobic portions dissolved in the micelle interior.
- Free fatty acids and monoglycerides are present in the inside of the micelle, essentially “solubilized” for subsequent absorption.

2. Formation of bile

- Bile is **produced continuously by hepatocytes**.
- Bile drains into the hepatic ducts and is stored and concentrated in the **gallbladder** for subsequent release. CCK tells the gallbladder that bile is needed to emulsify and absorb lipid in the duodenum. Thus CCK causes contraction of gallbladder and relaxation of the sphincter of Oddi to release bile. ACh also causes contraction of the gallbladder.

Digestion and Absorption

- Carbohydrates, proteins, and lipids are digested and absorbed in the small intestine.
- The surface area for absorption in the small intestine is greatly increased by the presence of the brush border.

A. Carbohydrates

1. Digestion of carbohydrates

- Only monosaccharides are absorbed. Carbohydrates must be digested to glucose, galactose, and fructose for absorption to proceed.
- a. **α -Amylases** (salivary and pancreatic) hydrolyze 1,4-glycosidic bonds in starch, yielding maltose, maltotriose, and α -dextrins.
- b. **Maltase, α -dextrinase, and sucrase** in the intestinal brush border then hydrolyze the oligosaccharides to glucose.
- c. **Lactase, trehalase, and sucrase** degrade their respective disaccharides to monosaccharides.
 - Lactase degrades lactose to glucose and galactose.
 - Trehalase degrades trehalose to glucose.
 - Sucrase degrades sucrose to glucose and fructose.

2. Absorption of carbohydrates

a. Glucose and galactose

- are transported from the intestinal lumen into the cells by a Na^+ -dependent cotransport (SGLT 1) in the luminal membrane. The sugar is transported “uphill” and Na^+ -dependent cotransport (SGLT-1) is transported “downhill.”
- are then transported from cell to blood by facilitated diffusion (GLUT 2).

- The Na^+/K^+ pump in the basolateral membrane keeps the intracellular $[\text{Na}^+]$ low, thus maintaining the Na^+ gradient across the luminal membrane.
- Poisoning the Na^+/K^+ pump inhibits glucose and galactose absorption by dissipating the Na^+ gradient.

b. Fructose

- is transported exclusively by **facilitated diffusion (GLUT-5)**; therefore, it cannot be absorbed against a concentration gradient.

3. Clinical disorders of carbohydrate absorption

- **Lactose intolerance** results from the absence of brush border lactase and, thus, the inability to hydrolyze lactose to glucose and galactose for absorption. Nonabsorbed lactose and H_2O remain in the lumen of the GI tract and cause **osmotic diarrhea**.

B. Proteins

1. Digestion of proteins

a. Endopeptidases

- degrade proteins by hydrolyzing interior peptide bonds.

b. Exopeptidases

- hydrolyze one amino acid at a time from the C terminus of proteins and peptides.

c. Pepsin

- is not essential for protein digestion.
- is secreted as pepsinogen by the chief cells of the stomach.
- Pepsinogen is activated to pepsin by gastric H^+ .
- The **optimum pH for pepsin is between 1 and 3**.
- When the pH is >5 , pepsin is denatured. Thus, in the intestine, as HCO_3^- is secreted in pancreatic fluids, duodenal pH increases and pepsin is inactivated.

d. Pancreatic proteases

- include trypsin, chymotrypsin, elastase, carboxypeptidase A, carboxypeptidase B.
- are secreted in inactive forms that are activated in the small intestine as follows:
 - (1) Trypsinogen is activated to **trypsin** by a brush border enzyme, enterokinase.
 - (2) Trypsin then converts chymotrypsinogen, proelastase, and procarboxypeptidase A and B to their active forms. (Even trypsinogen is converted to more trypsin by trypsin!)
 - (3) After their digestive work is complete, the pancreatic proteases degrade each other and are absorbed along with dietary proteins.

2. Absorption of proteins

- Digestive products of protein can be absorbed as amino acids, dipeptides, and tripeptides (in contrast to carbohydrates, which can only be absorbed as monosaccharides).

a. Free amino acids

- **Na^+ -dependent amino acid** cotransport occurs in the luminal membrane. It is analogous to the cotransporter for glucose and galactose.
- The amino acids are then transported from cell to blood by facilitated diffusion.
- There are **four separate carriers** for neutral, acidic, basic, and imino amino acids, respectively.

b. Dipeptides and tripeptides

- are absorbed faster than free amino acids.
- H^+ -dependent cotransport of dipeptides and tripeptides also occurs in the luminal membrane.
- After the dipeptides and tripeptides are transported into the intestinal cells, cytoplasmic peptidases hydrolyze them to amino acids.
- The amino acids are then transported from cell to blood by facilitated diffusion.

C. Lipids

1. Digestion of lipids

a. Stomach

- (1) In the stomach, **mixing** breaks lipids into droplets to increase the surface area for digestion by pancreatic enzymes.
- (2) **Lingual lipases** digest some of the ingested triglycerides to monoglycerides and fatty acids. However, most of the ingested lipids are digested in the intestine by pancreatic lipases.
- (3) **CCK slows gastric emptying**. Thus, delivery of lipids from the stomach to the duodenum is slowed to allow adequate time for digestion and absorption in the intestine.

b. Small intestine

- (1) **Bile acids** emulsify lipids in the small intestine, increasing the surface area for digestion.
- (2) **Pancreatic lipases** hydrolyze lipids to fatty acids, monoglycerides, cholesterol, and lysolecithin. The enzymes are pancreatic lipase, cholesterol ester hydrolase, and phospholipase A2.
- (3) The hydrophobic products of lipid digestion are solubilized in **micelles** by **bile acids**.

2. Absorption of lipids

- a. Micelles** bring the products of lipid digestion into contact with the absorptive surface of the intestinal cells. Then, **fatty acids, monoglycerides, and cholesterol diffuse across the luminal membrane into the cells**. Glycerol is hydrophilic and is not contained in the micelles.
- b.** In the intestinal cells, the products of lipid digestion are **re-esterified** to triglycerides, cholesterol ester, and phospholipids and, with apoproteins, form **chylomicrons**.
 - Lack of apoprotein B results in the inability to transport chylomicrons out of the intestinal cells and causes **abetalipoproteinemia**.
- c.** Chylomicrons are transported out of the intestinal cells by **exocytosis**. Because chylomicrons are too large to enter the capillaries, they are transferred to **lymph vessels** and are added to the bloodstream via the thoracic duct.

3. Malabsorption of lipids—steatorrhea

- can be caused by any of the following:
 - a. Pancreatic disease** (e.g., pancreatitis, cystic fibrosis), in which the pancreas cannot
 - synthesize adequate amounts of the enzymes (e.g., pancreatic lipase) needed for lipid digestion.
 - b. Hypersecretion of gastrin**, in which gastric H^+ secretion is increased and the duodenal pH is decreased. Low duodenal pH inactivates pancreatic lipase.
 - c. Ileal resection**, which leads to a depletion of the bile acid pool because the bile acids do not recirculate to the liver.
 - d. Bacterial overgrowth**, which may lead to deconjugation of bile acids and their “early” absorption in the upper small intestine. In this case, bile acids are not present throughout the small intestine to aid in lipid absorption.
 - e. Decreased number of intestinal cells** for lipid absorption (tropical sprue).
 - f. failure to synthesize apoprotein B**, which leads to the inability to form chylomicrons.

D. Absorption and secretion of electrolytes and H_2O

- Electrolytes and H_2O may cross intestinal epithelial cells by either cellular or paracellular (between cells) routes.
- **Tight junctions** attach the epithelial cells to one another at the luminal membrane.
- The permeability of the tight junctions varies with the type of epithelium. A “**tight**” (impermeable) epithelium is the colon. “**Leaky**” (permeable) epithelia are the small intestine and gallbladder.

1. Absorption of $NaCl$

- a.** Na^+ moves into the intestinal cells, across the luminal membrane, and down its electrochemical gradient by the following mechanisms:
 - (1) Passive diffusion (through Na^+ channels)
 - (2) Na^+ –glucose or Na^+ –amino acid cotransport
 - (3) Na^+ – Cl^- cotransport
 - (4) Na^+ – H^+ exchange

- In the **small intestine**, Na^+ -glucose cotransport, Na^+ -amino acid cotransport, and Na^+ - H^+ exchange mechanisms are most important. These cotransport and exchange mechanisms are similar to those in the renal proximal tubule.
- In the **colon**, passive diffusion via Na^+ channels is most important. The Na^+ channels of the colon are similar to those in the renal distal tubule and are stimulated by **aldosterone**.
- b. Na^+ is pumped out of the cell against its electrochemical gradient by the Na^+ - K^+ pump in the basolateral membranes.
- c. Cl^- absorption accompanies Na^+ absorption throughout the GI tract by the following mechanisms:
 - (1) Passive diffusion by a paracellular route
 - (2) Na^+ - Cl^- cotransport
 - (3) Cl^- - HCO_3^- exchange

2. Absorption and secretion of K^+

- a. Dietary K^+ is **absorbed in the small intestine** by passive diffusion via a paracellular route.
- b. K^+ is actively **secreted in the colon** by a mechanism similar to that for K^+ secretion in the renal distal tubule.
 - As in the distal tubule, K^+ secretion in the colon is stimulated by **aldosterone**.
 - In **diarrhea**, K^+ secretion by the colon is increased because of a flow rate-dependent mechanism similar to that in the renal distal tubule. Excessive loss of K^+ in diarrheal fluid causes **hypokalemia**.

3. Absorption of H_2O

- is secondary to solute absorption.
- is **isosmotic in the small intestine and gallbladder**. The mechanism for coupling solute and water absorption in these epithelia is the same as that in the **renal proximal tubule**.
- In the **colon**, H_2O permeability is much lower than in the small intestine, and feces may be hypertonic.

4. Secretion of electrolytes and H_2O by the intestine

- The GI tract also secretes electrolytes from blood to lumen.
- The secretory mechanisms are located in the **crypts**. The absorptive mechanisms are located in the villi.
- a. **Cl^- is the primary ion secreted** into the intestinal lumen. It is transported through Cl^- channels in the luminal membrane that are regulated by **cAMP**.
- b. Na^+ is secreted into the lumen by passively following Cl^- . H_2O follows NaCl to maintain isosmotic conditions.
- c. ***Vibrio cholerae* (cholera toxin)** causes diarrhea by stimulating Cl^- secretion.
 - Cholera toxin catalyzes **adenosine diphosphate (ADP) ribosylation** of the α_s subunit of the Gs protein coupled to adenylyl cyclase, permanently activating it.
 - Intracellular cAMP increases; as a result, **Cl^- channels** in the luminal membrane open.
 - Na^+ and H_2O follow Cl^- into the lumen and lead to **secretory diarrhea**.
 - Some strains of ***Escherichia coli*** cause diarrhea by a similar mechanism.

E. Absorption of other substances

1. Vitamins

- a. **Fat-soluble vitamins** (A, D, E, and K) are incorporated into micelles and absorbed along with other lipids.
- b. Most **water-soluble vitamins** are absorbed by Na^+ -dependent cotransport mechanisms.
- c. **Vitamin B₁₂ is absorbed in the ileum** and requires **intrinsic factor**.
 - The vitamin B₁₂-intrinsic factor complex binds to a receptor on the ileal cells and is absorbed.
 - **Gastrectomy** results in the loss of gastric parietal cells, which are the source of intrinsic factor. Injection of vitamin B₁₂ is required to prevent **pernicious anemia**.

2. Calcium

- absorption in the small intestine depends on the presence of adequate amounts of the active form of vitamin D, **1,25-dihydroxycholecalciferol**, which is produced in the kidney. 1,25-dihydroxycholecalciferol induces the synthesis of an intestinal Ca^{2+} -binding protein, **calbindin D-28K**.
- Vitamin D deficiency or chronic renal failure results in inadequate intestinal Ca^{2+} absorption, causing **rickets** in children and **osteomalacia** in adults.

3. Iron

- is absorbed as **heme iron** (iron bound to hemoglobin or myoglobin) or as free Fe^{2+} . In the intestinal cells, “heme iron” is degraded and free Fe^{2+} is released. The free Fe^{2+} binds to apoferritin and is transported into the blood.
- Free Fe^{2+} circulates in the **blood bound to transferrin**, which transports it from the small intestine to its storage sites in the liver, and from the liver to the bone marrow for the synthesis of hemoglobin.
- Iron deficiency is the most common cause of anemia.

Biochemistry

Carbohydrates

Carbohydrates (CHOs) are one of the three macronutrients (+ lipid + proteins) composed of carbon, hydrogen, oxygen. They can be simple sugars as monosaccharides (Glucose, fructose, mannose, galactose) and disaccharides (sucrose, Lactose, Maltose, Trehalose). They can also be complex CHOs (polysaccharides) such as starch, glycogen, cellulose, fibers, etc. CHO foods provide **4 kcal per gram**

Starch: storage form of CHOs in plants. It includes amylose and amylopectin (branched). Sources include: grains, legumes, and tubers. Glycogen: storage form of CHOs in animals; stored in the liver and muscles. Fibers: is the non-digestible part of plants like cellulose. Soluble Fibers: Dissolve in water, are viscous and gel-forming, associated with risk reduction of cardiovascular disease and Type 2-DM. Examples: pectin, gum, mucilage, are found in citrus fruits, berries, oats, beans. Insoluble Fibers: do not dissolve in water, are non-viscous, promote regular bowel movements. Examples: lignins, cellulose, hemicelluloses. Fibers differ from starch b/c the glucose units are connected by β -1-4 linkages; remember starch contains glucose units that are connected by α - 1-4 linkages. Amylase only works on the α 1-4 bonds, not β 1-4 bonds.

Recommended Dietary Allowance (RDA) for CHOs is 130 grams/day just to supply the brain with glucose. Acceptable Macronutrient Distribution Range (AMDR) for carbohydrates: 45-65% of daily calorie intake

CHO and Glycemic index

Glycemic Index (GI): is a measure of how quickly or slowly a CHO food is digested, absorbed and increases blood glucose level (measure of how quickly these foods make your blood glucose levels rise after eating them). Available CHOs: are CHOs that get readily digested, absorbed and metabolized by our body. These CHOs have a much greater impact on our blood sugar level than CHOs in general. We rank our CHO foods as being low, medium, or high in their GI value

Low GI	Medium GI	High GI
0 - 55	56 - 69	70 or greater

- **Higher GI CHO**s are high GI foods and drinks. They Increase blood glucose levels more **quickly**. These need to be chosen less often. **Lower GI CHO**s: increase blood glucose levels more **slowly**. These are the preferred choice.

Deficiency Diseases

Nutrient	Deficiency
Vitamin A (retinol)	Night blindness, dry skin, corneal degeneration
Vitamin B ₁ (thiamine)	Impaired glucose breakdown → ATP depletion Wernicke-Korsakoff syndrome and Beriberi Most common cause: chronic alcoholism
Vitamin B ₂ (riboflavin)	Cheilosis (inflammation of lips, scaling and fissures at the corners of the mouth), Corneal vascularization (2Cs of B ₂)
Vitamin B ₃ (niacin)	Diarrhea, Dermatitis and Dementia (3Ds of B ₃ leading to another D – i.e. death)
Vitamin B ₅ (pantothenic acid)	Dermatitis, enteritis, alopecia, adrenal insufficiency
Vitamin B ₆ (pyridoxine)	Convulsions, hyperirritability, peripheral neuropathy (deficiency is inducible by isoniazid and contraceptives), sideroblastic anemias due to impaired Hb synthesis and iron excess
Vitamin B ₇	Relatively rare; dermatitis, alopecia, enteritis Caused by antibiotic use or excessive ingestion of raw egg whites.
Vitamin B ₉ (folate)	Macrocytic, megaloblastic anemia; seen in alcoholism and pregnancy
Vitamin B ₁₂ (cobalamin)	Macrocytic, megaloblastic anemia, degeneration of dorsal columns, lateral corticospinal tracts, and spinocerebellar tracts due to abnormal myelin.
Vitamin C (ascorbic acid)	Scurvy (swollen gums, bruising, petechial hemorrhages, anemia, poor wound healing, weakened immune response, collagen synthesis defects)
Vitamin D (1,25-dihydroxy vitamin D, calcitriol)	Rickets in children (bone deformity and pain), osteomalacia in adults (bone pain and muscle weakness), hypercalcemic tetany <i>(cont'd)</i>

Vitamin E (tocopherol)	Hemolytic anemia (because Vitamin E is a major antioxidant), muscle weakness
Vitamin K (phylloquinone)	Neonatal hemorrhage
Zinc	Delayed wound healing, hypogonadism

Malnutrition – Kwashiorkor and Marasmus

- **Kwashiorkor** – a protein malnutrition resulting skin lesions, **edema** due to ↓ plasma oncotic pressure, **liver** malfunction (**fatty change** due to ↓ apoprotein synthesis), and anemia. Clinical picture is a small child with swollen abdomen.
- **Marasmus** – total calorie malnutrition resulting in tissue and **Muscle wasting**, loss of subcutaneous fat, and variable edema.

Metabolic Disorders

Metabolic disorder/deficiency	Deficiency	Features
Glucose, Fructose, Galactose and Lactose Metabolism disorders		
Pyruvate dehydrogenase complex (PDH) deficiency	PDH (X-linked)	Causes buildup of pyruvate that gets shunted by LDH (resulting lactic acidosis)
Glucose-6-phosphate dehydrogenase (G6PD) deficiency	G6PD (X-linked); the most common human enzyme deficiency	Hemolytic anemia (b/c no NADPH to keep glutathione reduced; Heinz bodies)
Essential fructosuria	Defect in fructokinase	Fructose in urine (very mild)
Fructose intolerance	Defect in aldolase B ¹¹ (fructose-1-phosphate accumulates in liver resulting symptoms)	Reducing sugars accumulate in urine, hypoglycemia, jaundice, cirrhosis, vomiting.
Galactokinase deficiency	Galactokinase (Galactitol accumulates)	Infantile cataracts
Classic galactosemia	Galactose-1-phosphate uridylyltransferase (galactitol accumulates in lens & liver)	Cataracts, jaundice, hepatomegaly
Lactose intolerance	Lactase deficiency	Bloating, flatulence, diarrhea
Amino acid Metabolism Disorders		
Hyperammonemia	Urea cycle enzyme deficiencies	Ammonia intoxication – tremor, cerebral edema (cont'd)

¹¹ Remember Aldolase A is in glycolysis (converts FBP (fructose 1,6 bisphosphate) to DHAP or G3P.

Phenylketonuria	Due to ↓phenylalanine hydroxylase or TH4 cofactor, tyrosine becomes essential	↑Phenylalanine – aromatic body odor (apple fruity smell)
Maple syrup urine disease	Blocked degradation of branched amino acids (Isoleucine, Leucine and Valine) due to ↓ α -ketoacid Dehydrogenase deficiency	Severe CNS defects; ↑ α -ketoacids in the blood
Alkaptonuria	Homogentisate oxidase deficiency	Debilitating arthralgias
Homocystinuria	Cystathionine synthase or homocysteine methyl transferase deficiency	Both forms result excess homocysteine – osteoporosis, kyphosis, atherosclerosis (MI and stroke), intellectual disability
Cystinuria	Hereditary defect of renal PCT and intestinal amino acid transporter that prevents reabsorption of Cysteine, Ornithine, Lysine and Arginine (COLA)	Cysteine stones
Glycogen Storage Disorders		
Von Gierke disease (type I glycogen storage disease)	Glucose-6-phosphatase	Severe fasting hypoglycemia, ↑↑ glycogen in liver, hepatomegaly, gout
Pompe disease (type II G.S dx)	Lysosomal α -1,4-glucosidase deficiency	Heart, liver and muscle damage
Cori disease (type III)	Debranching enzyme (α -1,6-glucosidase)	Milder form of type I (mild hypoglycemia...)
McArdle disease (type V)	Muscle glycogen phosphorylase deficiency	↑glycogen in muscle, painful muscle cramps with strenuous exercise esp. at onset of exercise
Lysosomal Storage Diseases		
Fabry Disease	α -galactosidase A	Peripheral neuropathy of hands and feet, CV, renal dx
Gaucher's disease	Glucocerebrosidase (β -glucosidase)	Hepatosplenomegaly, pancytopenia, osteoporosis

(cont'd)

Neimann-Pick disease	Sphingomyelinase	Progressive neurodegeneration, hepatosplenomegaly, foam cells – “cherry red spots on macula”
Tay-Sachs disease	Hexosaminidase A	Progressive neurodegeneration, developmental delay, “cherry red spots on macula” but no hepatosplenomegaly (vs. Neimann Picks)
Muccopolysaccharidosis		
Hurler syndrome	α -L-iduronidase	Developmental delay, gargoylism, corneal clouding
Hunter syndrome	Iduronate sulfatase	Mild hurler + aggressive behavior, no corneal clouding
Fatty Acid Metabolism Disorder		
Medium chain acyl-CoA dehydrogenase deficiency	Medium chain acyl-CoA dehydrogenase	↓ability to break down fatty acids into Acetyl-CoA, hypoketotic hypoglycemia. vomiting, lethargy, seizures, coma, and liver dysfunction.
Familial Dyslipidemias		
Type-I (hyperchylomicronemia)	Lipoprotein lipase deficiency or altered apolipoprotein CII; Increased Blood Level of Chylomicrons, TG, cholesterol	Pancreatitis, hepatosplenomegaly, xanthomas, (no ↑risk for atherosclerosis)
Type-IIa (familial hypercholesterolemia)	Absent or defective LDL receptors; increased blood levels of LDL and cholesterol	Causes accelerated atherosclerosis (may have MI before 20yrs), Xanthomas on Achilles tendon
Type IV (hypertriglyceridemia)	Hepatic overproduction of VLDL; increased blood levels of VLDL and TGs.	Hypertriglyceridemia can cause pancreatitis.

Microbiology

Microbial Causes of Diarrheal disease

Note that Diarrhea, the passage of abnormal amounts of fluid or semisolid fecal matter, can be mediated by a number of mechanisms (not just by micro-organisms). Osmotic diarrhea results when unabsorbed solutes increase intraluminal oncotic pressure, causing an outpouring of water. Active ion secretion causing obligatory water loss is termed secretory diarrhea. Sloughing of colonic mucosa, caused by inflammation and necrosis, often as a result of infection, causes an exudative form of diarrhea which will be discussed.

- i. Bacterial causes of diarrhea

ii. Viral causes of diarrhea

iii. Protozoal causes of diarrhea (causes persistent diarrhea)

Bacterial causes of diarrhea		
Infectious agent	Clinical Features	Notes
<i>Shigella</i>	Shiga toxin causes bloody diarrhea, mild to severe, 1-2 weeks in duration; fever; requires small amount of MOs (~100 organisms)	Fecal leukocytes and stool culture necessary for diagnosis
<i>Salmonella</i>	Bloody diarrhea, nausea, motile ; fever; requires large amount of MOs (~100 organisms) ingestion	Commonly acquired from eggs, poultry , diagnosis based on stool culture, increased susceptibility in immunocompromised .
<i>Campylobacter jejuni</i>	Bloody diarrhea, fever, crampy abdominal pain, self-limited	Spiral shaped, oxidase + bacteria
<i>Vibrio cholerae</i>	Watery diarrhea (rice water stools), vomiting, and dehydration occurs after 12-48hrs of incubation	Caused by toxin , comma-shaped bacteria with flagellum; associated with poor living conditions
<i>Clostridium difficile</i>	Watery diarrhea caused by antibiotic-induced suppression of colonic flora and <i>C. difficile</i> overgrowth, pseudo-membranes on the colonic mucosa (hence aka pseudomembranous colitis)	Exotoxin mediated, termed pseudomembranous colitis because of the false membranes created on the colon by the bacterial infection; Tx: Metronidazole or vancomycin
Enterotoxigenic <i>Escherichia coli</i> (ETEC) – (Traveler's diarrhea)	Watery diarrhea; 3-6 days' duration, occasional fever and vomiting	Antibiotics reduce duration of infection to 1-2 days
Enterohemorrhagic <i>E. coli</i> (O157:H7)	Shiga-like toxin causes bloody diarrhea	Typically foodborne transmission (e.g. undercooked hamburger)
<i>Yersinia enterocolitica</i>	Bloody diarrhea, fever, nausea, cramps	Transmitted by food or contaminated domestic animal feces; clinically indistinguishable from <i>Salmonella</i> or <i>Shigella</i> .

Viral Causes of Diarrhea		
Infectious Agent	Clinical Features	Notes
<i>Rotavirus</i>	Severe, dehydrating diarrhea; vomiting; low-grade fever	The MCC (most common cause) of infantile diarrhea
<i>Norwalk virus</i>	Mild diarrhea & vomiting	Affects both children & adults
<i>Adenovirus</i> (serotypes 40 & 41)	Diarrhea and moderate vomiting	Next to rotavirus – the 2 nd MCC of gastroenteritis in children

Protozoal Causes of Diarrhea		
Infectious agent	Clinical Features	Notes
<i>Entamoeba histolytica</i>	Bloody diarrhea, lower abdominal pain, may lead to dysentery with bloody and mucous stools per day.	Caused by ingestion of viable cysts via fecal-oral route (cont'd)

<i>Giardia lamblia</i>	Watery , foul-smelling diarrhea; nausea; anorexia; cramps lasting weeks to months	Fecal-oral transmission; often contracted while camping
<i>Cryptosporidium</i>	Watery diarrhea with large fluid loss; symptoms persist in immunocompromised patients; self-limited in healthy individuals	Immunocompromised patients (especially patients with AIDS); fecal-oral transmission of oocysts

Microbiology of food Poisoning

- Many GIT infections involve food as a vehicle of transmission. When a patient presents with **GIT symptoms** including **nausea, vomiting, abdominal pain, diarrhea** and **fever**, most clinicians consider a foodborne illness. However, patients with foodborne illness may present initially with other complaints such as neurologic symptoms (e.g. headaches, paralysis or tingling), hepatitis, and renal failure.
- The probable etiologic agent can usually be assessed from knowledge of the **incubation period, the food vehicle, and the clinical findings**. Food poisoning can be caused by either **infection** or **intoxications**.
- Classification of pathogens that causes food poisoning
 - Gram (+) cocci -- *S. aureus*
 - Gram (-) cocci – *salmonella, shigella*
 - Gram (+) rod -- *B. cereus* (aerobic); *C. botulinum* and *C. perfringens* (anaerobic)
 - Gram (-) rod -- *Vibrio parahaemolyticus*
- Salmonella, C. perfringens* and *S. aureus*** accounting for more than **70%** of those for which a microbial etiology.
- Intoxications**: have **shorter incubation periods** than infections and may involve **extra intestinal symptoms** (e.g. the neurologic damage in botulism).
- Etiologic agents include
 - S. aureus*** – ingestion of food containing the bacterial enterotoxin. Often contaminated by a food-handler, these foods tend to be protein-rich and/or salty (for example, egg, salad, cream pastry, or ham) and improperly refrigerated. The toxin stimulates the vomiting center in the brain by binding to neural receptors in the upper GIT. Symptoms such as **nausea, vomiting**, and **diarrhea** are acute following a short incubation period (**less than 6 hours**). It resolves in **less than 24 hrs.** without treatment.
 - B. cereus*** -- ingestion of toxin. Symptoms in **1 to 6 hrs.** Similar to Staphylococcal food poisoning.
 - C. botulinum*** -- ingestion of the preformed toxin in contaminated food (improperly canned food, foil-wrapped potatoes and meat pies held in restaurants). The disease is a **pure intoxication**. There is **no fever** or sign of **sepsis**. Initial symptoms include double vision, fixed pupils, dry mouth, dizziness, and constipation with **flaccid paralysis** within **12 – 72 hrs.** Recovery, which involves regeneration of the damaged nerves, is protracted, **lasting several weeks**
 - Chemicals** -- Includes heavy metals, monosodium glutamate, mushrooms, and various toxins of non-microbial origin
 - Infectious**: food poisoning does not differ from endemic diarrheal infections caused by the same species. The **length of the incubation period** and the **severity** of the symptoms are generally related to the **number of organisms ingested**.

- Etiologic agents include
- ***C. perfringens*** -- The occurrence of clinical symptoms requires a **large inoculum of 10^8** organisms or greater. Therefore, a typical episode of clostridial food poisoning involves **cooking that fails to inactivate spores**, followed by holding the food for several hours under conditions that allow **bacterial germination and several cycles of growth**. It secretes various exotoxins, enterotoxins, and hydrolytic enzymes that facilitate the disease process. Symptoms typically include **nausea, abdominal cramps and diarrhea**, occurring **8 to 18 hrs** after eating contaminated food. **Fever is absent and vomiting rare**. The attack is usually **self-limited**, with recovery within **1 to 2 day**.
- ***Salmonella*** -- is a **non-lactose fermenter**, is **motile** (like a salmon which is large fish) and **produces H_2S** . There are thousands of *Salmonella* serotypes, but clinically they are usually divided into three groups: ***Salmonella typhi***, ***Salmonella typhimurium***, and ***Salmonella enteritidis***. *Salmonella* differs from the other enterics because it lives in the GIT of animals **except *salmonella typhi*** which is carried only by humans. Transmitted by fecal–oral transmission. Salmonellosis is caused primarily by serovars *Enteritidis* and *Typhimurium* and is characterized by **nausea, vomiting, and diarrhea** (usually **non-bloody**) occurring within **6-48 hrs**. In non-immunocompromised patients, disease is generally **self-limiting (48 to 72 hrs)**.
- ***Shigella*** -- There are four species of *Shigella* (*dysenteriae*, *boydii*, *flexneri* and *sonnei* from most severe to mild) and all are **nonmotile**, has **no flagella**, **non-lactose fermenter** and does **not produce H_2S** . Humans are the only hosts for *Shigella*. It is typically spread from person to person, fecal-to-oral route occurs via fecally contaminated water and food. It produces shiga toxin similar to enteroinvasive *Escherichia coli* (EIEC) that invade intestinal epithelial cells which causes cell destruction and responsible for pathogenesis. It is the cause shigellosis (bacillary dysentery) commonly among young children. The resulting bacillary dysentery is characterized by **diarrhea with blood, mucus, and painful abdominal cramping** occurring **within 12-48 hrs**. Among non-immunocompromised populations, untreated dysentery commonly resolves in a **week** but may persist longer.
- ***Vibrio parahaemolyticus*** -- this organism is a marine bacterium that causes **watery diarrhea** after ingestion of uncooked seafood (Shellfish)

Diagnosis

- *Salmonella* and *Shigella* which are not normal members of the stool flora, establishing the diagnosis by isolating the causative organism in the **stool**
- *C. perfringens* or *S. aureus* food poisoning, investigation involves cultures of **vomit**, **stool** from several cases, and the **suspect food**
- Clinical manifestations may help you for diagnosis

Prevention

“Keep yourself from food poisoning by eating properly cooked & well refrigerated foods and making yourself clean especially during preparation for exit exam A. Going now and then to rest room may waste your time!” - Zaki

Viral Hepatitis

	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
Virus family	Picornavirus	Hepadnavirus	Flavivirus	Delta agent	Calcivirus
Viral morphology	ssRNA	dsDNA	ssRNA	ssRNA	ssRNA
Mode of transmission	Fecal-oral	Sexual, parenteral, transplacental	Same as Hep B	Same as B	Fecal-oral
Diagnostic test	IgM	HbsAg, anti-HbsAg, HbeAg, anti-HbcAg	Anti-HCV	Anti-sag	None
Severity	Mild	Moderate	Mild	Severe	Mild
Chronic infection	No	In infants and Immuno C. (90%)	80-90%	No, if alone	No
Carrier state	No	Yes	Yes	Yes	No
Hepatocellular carcinoma	No	Yes	Yes	No	No
Prophylaxis and treatment	Ig, vaccine	HB Ig, vaccine,	Interferon and ribavirin	HB Ig and vaccine	None
Notes	IP = 14-15d	Dane particle: is the viral DNA genome + DNA polymerase, HbcAg, HbeAg, HbsAg,	Most frequent cause of transfusion-mediated Hepatitis	Defective replication; requires coinfection with Hep B.	Severe among pregnant humans

Hepatitis B – Terminology and Markers

Abbreviation	Name and description
Dane particle	Component of infectious HBV (Hepatitis B virus – an enveloped dsDNA virus)
HBsAg	HB surface Antigen (found on the surface of HBV); positive during acute disease, continued presence indicates carrier state
HBsAb	Antibody to the surface antigen (HBsAg); presence means immunity
HBcAg	Antigen associated with core of HBV (located in the core)
HBcAb	Antibody to HBcAg; positive during windows period (hence the only marker during WP), presence of HBcAb indicates a recent disease
HBeAg	A second, different antigenic determinant on the HBV core; important indicator of transmissibility (means high infectivity and viral proliferation)
HBeAb	Antibody to e antigen indicates low transmissibility
Window period	The period b/n the end of detection of the HBsAg and the beginning of detection of HBsAb

Hepatitis B Serology

	HBsAg, HBeAg*, HBV-DNA	HBcAb, IgM	HBcAb, IgG	HBeAb	HBsAb
Acute infection	+	+	-	-	-
Window period	-	+/-	+	+	-
Prior infection	-	-	+	+	+
Immunization (cured status)	-	-	-	-	+
Chronic infection	+	-	+	+/-	-
Chronic active	+	-	+	+	-

HBeAg – correlates with viral proliferation and infectivity – if present in chronic infection indicates that the person has a potential of transmitting (i.e. chronic active infection)

Pathology

Diseases of Oral Cavity and Salivary glands

Diseases of Oral Cavity and Salivary glands		
Diseases	Definitions	
Aphthous Ulcers (canker sores)	Non-infectious ulceration of oral cavity, one of the preAIDS defining lesions with candidiasis and hairy leukoplakia (not AIDS defining tho)	Gross: Hyperemic ulcerations covered by a thin exudate with hyperemic rim (red border) Micro: initially PMN infiltrate then neutrophils (unlike other infections neutrophils come late)
Candidiasis (Oral thrush)	Normal flora that overgrows due to diabetes, AIDS, immunosuppression	Gross: superficial curdy-white inflammatory exudate Micro: membrane composed of fungi in fibrinopurulent exudate resting on inflammatory base
Gingivitis	Painless inflammation of gingiva around gingival sulci or cuff; due to bacterial plaque (<i>S. mutans</i> – <i>Group D Strep</i>)	Gross: erythematous & edematous gingiva adjacent to teeth; Micro: initially neutrophils, later T lymphocytes, finally, B lymph.
Herpes Simplex Virus lesions	Usu. HSV-1; becomes dormant in trigeminal and other local ganglia; most infected adults have <u>latent HSV-1</u> activated by URT infection, allergy,	Gross: cluster of vesicles around the vermilion border of both lips Micro: intra-nuclear viral inclusion bodies
Hairy leukoplakia	Associated with <i>HIV infection</i> (AIDS may appear in 2-3yrs) but actually due to EBV .	Gross: unlike thrush, the lesion can't be scrapped off. Micro: Cowdry type A intranuclear viral inclusions
Pyogenic granuloma	Most common tumor of the gingiva, it's more of a malformation (a form of hemangioma not truly neoplastic);	associated with gingivitis; <i>pyogenic</i> – is a misnomer (not pus-forming; not chronic disease).
Squamous papilloma	Common intraoral benign neoplasm, associated with HPV (low risk types – 6 & 11)	Gross: cauliflower-like surface projections; Micro: fibrovascular cores
Leukoplakia	Precancerous; white plaque like lesion, try to scrap off, won't come off	Micro: thickened white patches of epithelium (no dysplasia and/or cancer – rule them out)
Erythroplakia (dysplastic leukoplakia)	Red patch; high tendency of progressing to CA (SCC) than leukoplakia (2X)	Micro: highly atypical epithelial changes
SCC	Most common CA of oral cavity (95%); 90% - men; Causes: alcohol, tobacco, marijuana, betel nuts and pan (India)	Gross: initially leukoplakia – then ulcerates with necrosis; induration (hardening) is specific for invasion. Micro: verrucoid growth pattern; atypia at base.
Kaposi's Sarcoma	Associated with HIV; oral lesion may be the first sign of dx, caused by HHV-8 – human herpes virus type 8 (KSV)	Gross: lesions all over the body – very painful – because they follow dermatomes

(cont'd)

Sialadenitis	Inflammation of the salivary glands - Bacterial (rare – caused by Staph aureus), viral (very common – mumps causes parotitis) or autoimmune. Usually occurs b/c of obstruction by calculus or mineral stones (sialolithiasis)	Mumps also causes orchitis and oophritis (inflammation of testis and ovary)
Chronic sialadenitis	Aka lymphoepithelial sialadenitis (relatively common), associated with obstruction, Sjogren's syndrome (autoimmune destruction)	Micro: chronic lymphocytic inflammation
Sialolithiasis	Stones (calculi) within the salivary ducts; most common with the submandibular gland	
Epithelial (myoepithelial) tumors	Most are in parotid gland; sublingual tumors are rare (parotid, in general is MC site for salivary gland tumors)	
Pleomorphic adenoma	Benign tumors that consist mixture of ductal (epithelial) and myoepithelial cells, and therefore they show both epithelial and mesenchymal differentiation	Examples include: Warthin's tumor (heterotypic salivary gland tissue in lymph node), Oncocytoma, basal cell adenoma, ductal papilloma
Mucoepidermoid carcinoma	MC malignant salivary gland tumor	

Non-neoplastic and Neoplastic diseases of the esophagus

Non-neoplastic and neoplastic diseases of the esophagus		
Mallory Weiss Syndrome (tear)	Laceration of the GE (gastroesophageal) junction; usually due to severe retching (vomiting) and associated with alcoholism	Gross: a longitudinal tear at GE junction; upper GI bleeding
Esophageal varices	Bleeding from esophageal varices due to portal HTN	Gross: dilated tortuous vessels,
Esophagitis	Defined as epithelial damage and inflammation; MCC is gastroesophageal reflux (reflux of gastric contents into lower esophagus); infection: rare	Micro: Erosion, laceration, exudative
GERD (Gastroesophageal reflux disease)	Most common cause of esophagitis; Usu. In the elderly . Causes: CNS depressants, hypothyroidism, alcohol, tobacco, pregnancy	Long term consequences: bleeding, stricture, Barret's esophagus, ulcer
Infectious esophagitis	Usu. Restricted to immunocompromised patients or diabetic or those on antibiotic	Most common Cause: Candida (This is AIDS defining) – when it gets to esophagus.
Barret's esophagus	Distal squamous mucosa is replaced by metaplastic columnar epithelium in response to prolonged injury; Cause: GERD; irreversible	Major risk factor for esophageal adenocarcinoma Micro: presence of goblet cells (highly distinguishing); b/c goblet cells are of columnar epi
SCC of the esophagus	Most common type of esophageal CA; Most common site is mid-esophagus,	Gross: deep irregular ulcers with nodular margins, (cont'd)

Adenocarcinoma	Increasing incidence for unknown reasons; arises in a setting of Barret's esophagus	Gross: usu. Distal esophagus with invasion of gastric cardia Micro: moderate to well differentiated; mucin producing

Non-neoplastic and Neoplastic diseases of the stomach

Non-neoplastic and neoplastic diseases of the stomach		
Active gastritis	Acute mucosal inflammatory process, usually transient; associated with heavy use of NSAIDs, excessive alcohol use, heavy smoking, CA chemotherapy, bile reflux, systemic infections, severe stress, acid ingestion/as part of suicidal attempts	Micro: mild edema of lamina propria, intact epithelium, scattered neutrophils; severe erosion (loss of epithelium above muscularis mucosa) and hemorrhage in mucosa.
Autoimmune gastritis	Is a type A (fundal) chronic gastritis; rare as compared to type B (antral) gastritis that is due to H. Pylori	Associated with hypochlorhydria/ achlorhydria due to parietal cell loss by anti-parietal cell Ab; also the most common cause of pernicious anemia – due to anti-intrinsic factor Antibody.
H. Pylori gastritis (aka antral or type B gastritis)	Associated with duodenal ulcer and gastric peptic ulcer, epithelial changes may become dysplastic & progress to CA	Gross: thin, smooth mucosa (without rugae), prominent submucosal vessels.
Peptic Ulcer disease (PUD)	Defined as a breach in muscularis mucosa of the GI tract; can be chronic, usually solitary; Causes: mucosal injury due to H. Pylori infection, NSAID use, ZE syndrome (Gastronema)	Acute PUD: due to NSAID use Chronic PUD: coexisting duodenal ulcer Both lead to hemorrhage, perforation; NO heaped margins in both cases (no piling up of the mucosa)
Adenoma	Represents 5% of polyp lesions in the stomach (less common); precursor lesion of gastric adenocarcinoma	Gross: large, single, occur in antrum, can be sessile or pedunculated. Micro: polypoid projections of dysplastic epithelium
Hyperplastic polyp	Aka Inflammatory, regenerative polyps, much common than adenoma (the most common polyp); this one is non-neoplastic	Gross: small , sessile, multiple, central umbilication is very common & characteristic Micro: no dysplasia (vs adenoma)
Carcinoma	Usually asymptomatic until late, depth of invasion is important prognostic factor; High incidence in Japan; metastasis to supraclavicular nodes	Death due to widespread metastasis; results Krukenberg's tumor if it metastasizes to the ovary
Intestinal type adenocarcinoma	Bulky tumors composed of glandular structures; risk factors: diet (smoked foods, nitrosamines), chronic gastritis (H.	Gross: healed up margin; most commonly involves the lesser curvature of antrum <i>(cont'd)</i>

	pylori or autoimmune ... lead to intestinal metaplasia)	Micro: neoplastic intestinal glands resembling colonic adenocarcinoma
Diffuse type adenocarcinoma	Aka linitis plastica or signet ring adenocarcinoma ; infiltrative growth of poorly differentiated malignant cells (signet ring cells)	Gross: thickened, rigid, leather bottle like stomach May cause pyloric obstruction Not associated with diet or intestinal metaplasia Micro: usu. Do NOT form glands, gastric mucus cells (signet ring cells – mucin pushes their nucleus – making them look like a ring)

Common Diseases of the Intestines

Common Diseases of the Intestines		
Duodenal ulcers	Associated with Helicobacter Pylori (in most cases)	
Intestinal obstruction	Can be mechanical or functional; 80% of mechanical intestinal obstruction is caused by hernias (esp. weight lifters), intestinal adhesions (MCC of obstruction after surgery), intussusception (occurs when the terminal ileum intussuscepts or gets into the cecum), & volvulus (twisting of the colon). In all cases, there will be a colicky pain	The diagrams show: 1. Herniation: A loop of intestine protruding through a weak spot in the abdominal wall. 2. Adhesions: Fibrous bands (scar tissue) pulling on the intestines. 3. Volvulus: A segment of the large intestine twisting around its own axis. 4. Intussusception: A segment of the small intestine telescoping into the next segment.
Ischemic Bowel disease	Atherosclerosis of celiac artery and SMA resulting in ischemia of the bowel. There are 3 types: mucosal, mural (mucosal + submucosal) and transmural infarction	Usually affects watershed areas (splenic flexure or recto-sigmoid junction)
Hemorrhoids	Occur secondary to persistently elevated venous pressure within the hemorrhoidal plexus; Internal vs. external	Internal hemorrhoids bleed and are painless; while external hemorrhoids thrombose and are painful
Malabsorption	Defective malabsorption of everything (macronutrients, vitamins, electrolytes, minerals and water)	Steatorrhea : excessive fecal fat, bulky, frothy, greasy, yellow, or clay-colored stools
Celiac disease	Immune mediated damage of small bowel villi due to gluten exposure (Gluten – found in wheat; its most pathologic component is gliadin)	Helper T cells mediate damage; Terminal digestion and transepithelial transport intact
Whipple disease	Systemic damage characterized by macrophages loaded with Tropheryma whippelli organisms; partially destroyed	Classic site of involvement is small bowel lamina propria (cont'd)

	organisms are present in macrophage lysosomes		
Abetalipoproteinemia	Autosomal deficiency of apoprotein B-48 and B-HW		Malabsorption – due to defective chylomicron formation (requires H-4S); absent VLDL and LDL (require B-48)
Carcinoid tumor	Malignant proliferation of neuroendocrine cells; low grade malignancy; often secretes serotonin ; small bowel is most common site of origin		5-HIAA (5-hydroxy indole-acetic acid), a breakdown product of serotonin can be detected in urine.
Inflammatory Bowel Disease	Crohn disease Location – “small int., colon, mouth-anus” Histologic findings – full thickness inflammation; Granulomas Gross: Cobblestone appearance , wall thickening with narrowed lumen, skipped areas , Fistulas Risk of malignancy – small Recurrence after surgery - common Treated with infliximab only (= an anti-inflammatory-monoclonal Ab that binds to TNF and inhibits inflammation)		Ulcerative Colitis Location- colon and rectum Histologic findings – mucosal inflammation; crypt abscess Gross: pseudo polyps, widened lumen, toxic megacolon , diffuse Risk of malignancy – high Recurrence after surgery – No Treated with sulfasalazine ¹² (anti-inflammatory; activated by colonic bugs) + infliximab
Diverticulosis	Outpouchings of colonic mucosa and submucosa; most common cause of lower GI bleeding; most common site for diverticulosis, cancer, polyps in the whole GIT is the sigmoid colon		Resembles a left-side appendicitis
Juvenile (hamartomatous) Polyp	A hamartomatous polyp (not precancerous), mostly occur in the rectum, undergo torsion – cause rectal bleeding		Mostly occurs in kids
Neoplastic Polyp	Tubular adenoma	Tubulovillous adenoma	Villous adenoma
	- Usually Benign - Multiple - Pedunculated - Most common polyp	- Greater potential of malignancy than tubular adenoma - Morphologically shares features of both tubular & villous adenoma	- Highly malignant - Sessile tumors, fingerlike projections
Familial adenomatous Polyp	An autosomal dominant disorder; mutation of APC gene, ras, and P53 are known		presence of >100 polyps confirm FAP; Tacot's sequence (triad) – presence of FAP with gliomas and medulloblastoma)
Colorectal adenocarcinoma	Arises from colonic or rectal mucosa; can develop Left sided colon CA vs right side colon CA		Left side bleeds, right side obstructs
Appendicitis	Inflammation of the abdomen; pain at McBurney's point		False positive rate is 20% Perforation (rupture) → peritonitis → Dissemination

¹² Sulfasalazine = sulfa pyridine (antibacterial) + mesalamine (anti-inflammatory)

Diseases of the Liver, gallbladder, and pancreas

Neoplastic and non-neoplastic diseases of the liver		
Jaundice	Yellow discoloration of skin	earliest sign is scleral icterus (yellow discoloration of sclera)
Viral Hepatitis	Inflammation of liver parenchyma; mostly caused by Hepatitis B virus	Acute or chronic hepatitis
Cirrhosis	End stage liver damage characterized by disruption of hepatic parenchyma by fibrosis	Result in Portal HTN (due to fibrosis), leading to ascites, hypersplenism, portosystemic shunts (esophageal varices, hemorrhoids and caput medusa)
Alcohol related liver disease	Damage to the hepatic parenchyma due to consumption of alcohol; fatty liver is accumulation of fat in hepatocytes; alcohol hepatitis results from chemical injury to hepatocytes (acetaldehyde mediates damage)	Characterized by Mallory bodies (damaged cytokeratin filaments); presents with painful hepatosplenomegaly and elevated liver enzymes (ALT>AST); cirrhosis is a late complication → death
Non-alcoholic fatty liver disease	Fatty change, hepatitis and/or cirrhosis that develop without exposure to alcohol;	associated with obesity; ALT>AST
Hemochromatosis	Excess body iron leading to deposition in tissues (hemosiderosis) and organ damage (hemochromatosis)	Tissue damage is mediated by generation of free radicals; due to autosomal recessive defect in iron absorption (primary) or chronic transfusions (secondary); risk factor for HCC
Primary Biliary cirrhosis	Autoimmune granulomatous destruction of intrahepatic ducts; etiology unknown, anti-mitochondrial antibody is present	Presents with features of obstructive jaundice; cirrhosis is a late complication
Primary Sclerosing cholangitis	Inflammation and fibrosis of intrahepatic and extrahepatic ducts	Periductal fibrosis with onion skin appearance ; increased risk for cholangiocarcinoma
Reye's Syndrome	Fulminant Hepatitis and encephalopathy in children with viral illnesses who take aspirin	Likely related to mitochondrial damage of hepatocytes
Hepatic adenoma	Benign tumor of hepatocytes; associated with oral contraceptive use (regresses upon cessation of the drug)	Risk of rupture and intraperitoneal bleeding, esp. during pregnancy
Hepatocellular carcinoma	Malignant tumor of hepatocytes; increased risk for Budd Chiari syndrome (liver infection 2 ^o to hepatic vein obstruction); serum tumor marker is AFP	Risk factors include: chronic hepatitis, cirrhosis, anatoxins from <i>Aspergillus</i> (induce P53 mutation); symptoms appear late b/c are masked by cirrhosis

(cont'd)

Metastasis to the liver	More common than primary liver tumors; most commonly from colon , pancreas, lung and breast	Results multiple nodules in the liver
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Neoplastic and non-neoplastic diseases of the gallbladder		
Cholelithiasis (Gallstones)	Solid, round stones in the gallbladder; due to precipitation of cholesterol or bilirubin in the bile; cholesterol stones are more common	Gallstones are usually asymptomatic; complications include biliary colic, acute and chronic cholecystitis, ascending cholangitis, gallstone ileus (if small can be lodged in ileum)
Biliary Colic	Coming and going RUQ pain	Due to gallbladder contracting against stone lodged in the cystic duct; common bile duct obstruction may cause pancreatitis or obstructive jaundice
Acute cholecystitis	Acute inflammation of gallbladder wall; impacted stone in the cystic duct results dilation with pressure ischemia – bacterial overgrowth, inflammation	Presents with RUQ pain, often radiating to the right scapula; Risk of rupture if untreated,
Chronic Cholecystitis	Chronic inflammation of the gallbladder; due to chemical irritation from long standing cholelithiasis, with/without acute cholecystitis	Characterized by herniation of gallbladder mucosa into the muscular wall (Rokitansky-Aschoff sinus)
Ascending cholangitis	Bacterial infection of the bile duct; usually die to ascending infection with enteric gram – bacteria	Increased incidence with choledocholithiasis (stone in biliary ducts)
Gallstone ileus	Gallstone enters and obstructs the small bowel	
Gallbladder carcinoma	Adenocarcinoma arising from the glandular epithelium that lines the bladder wall	Gallstones are the major risk factor; Poor prognosis

Neoplastic and non-neoplastic diseases of the pancreas		
Acute pancreatitis	Inflammation and hemorrhage of the pancreas – due to auto digestion of pancreatic parenchyma by pancreatic enzymes (premature activation of trypsin leads to activation of other pancreatic enzymes)	Results liquefactive necrosis of the pancreas and fat necrosis of peri-pancreatic fat (fat around the pancreas); commonly due to alcohol and gallstones; clinical feature: epigastric pain radiating to the back
Chronic pancreatitis	Fibrosis of pancreatic parenchyma, most often secondary to recurrent acute	Clinical features: epigastric pain radiating towards the back; (cont'd)

	pancreatitis – most commonly due to alcohol (adults) and cystic fibrosis (children) ; however, many cases are idiopathic.	pancreatic insufficiency – results malabsorption with steatorrhea and fat-soluble vitamin deficiencies; dystrophic calcification of pancreatic parenchyma on imaging; increased risk for pancreatic CA
Pancreatic carcinoma	Adenocarcinoma arising from the pancreatic ducts; most commonly seen with the elderly; major risk factors are smoking and chronic pancreatitis	Clinical Features (usu. Occur late): epigastric abdominal pain and weight loss; obstructive jaundice with pale stools and palpable bladder with tumors that arise from the head of the pancreas (most common location); pancreatitis; treatment: Whipple's procedure; very poor prognosis

Inflammation of the Peritoneum

Inflammation of the peritoneum (Peritonitis)		
Peritonitis	Can be localized or diffuse (in location), acute or chronic (in history), infectious or aseptic in pathogenesis	
Acute peritonitis	Is most often infectious and is usually related to perforated viscus; aka secondary peritonitis	Due to: perforation of the bowel (by appendicitis, diverticulosis..), perforation of other organ (pancreatitis, cholecystitis...)
Infectious peritonitis	Aka primary peritonitis; when no other intra-abdominal source is identified;	occurs in a setting of ascites and liver cirrhosis
Aseptic peritonitis	Due to peritoneal irritation by abnormal presence of physiologic fluids or sterile foreign bodies	Can also occur as a complication of rare systemic disease
Localized peritonitis	Limited to an area of inflammation;	Can occur due to uncomplicated appendicitis; diverticulitis
Generalized	Widespread inflammation and diffuse abdominal tenderness	Many causes – complicated appendicitis, pancreatitis...

Pharmacology

Antipeptic Ulcer Drugs

Antipeptic ulcer drug	Mechanism; Uses	Side effects
H ₂ Antagonists: cimetidine, ranitidine, famotidine	Used in PUD (less effective than PPIs), GERD, ZE syndrome	Cimetidine is a major P450 inhibitor; cimetidine: ↓androgens → gynecomastia and ↓libido <i>(cont'd)</i>

Proton pump inhibitors (PPIs) – Omeprazole and other –prazoles	MOA: irreversible, direct inhibitors of the proton pump (K ⁺ /H ⁺ transport) in the gastric parietal cell; Uses: more effective than H ₂ blockers, effective in GERD and ZE; component of eradication regimen for <i>H. pylori</i>	
Misoprostol	MOA: PGE ₁ analogue which is cytoprotective; ↑mucus, ↓HCl Uses: selective use in NSAID induced GI ulcers	
Sucralfate	MOA: polymerizes on GI luminal surface to form a protective gel-like coating ulcer beds (requires acid pH so antacids may interfere); Uses: ↑healing and ↓ulcer recurrence	
Bismuth Subsalicylate	MOA: binds selectively to ulcer, coating and protecting it from acid and pepsin; combined with metronidazole, & tetracycline to eradicate <i>H. Pylori</i> (BMT regimen)	
Antacids: Al(OH) ₃ , Mg (OH) ₂ , CaCO ₃	MOA: bases that neutralize protons in gut lumen	Side effects: constipation (Al ³⁺) and diarrhea (Mg ²⁺)

Purgatives

- Drugs used for constipation (aka laxatives)
- **Methylcellulose**: bulk forming laxative – dietary fibers
- **Lactulose**: Osmotic laxative; other use: also decreases ammonia in hepatic encephalopathy
- **Magnesium hydroxide** (milk of magnesia) – osmotic laxative
- **Magnesium sulfate, magnesium citrate** – osmotic laxatives
- **Docusate** – stool softener – by emulsifying stool; makes the passage easier.
- **Metoclopramide** – prokinetic, antiemetic and purgative agent; **diabetic and post-operative gastroparesis**.

Antidiarrheal drugs

- Drugs used in diarrhea
- **Loperamide**: Class: an opioid agonist; oral antidiarrheal

Antiemetics

- Drugs used for nausea and vomiting, include:
 - 5HT₃ (a serotonin receptor) antagonists: **ondansetron** (used in CA chemotherapy; ADR: QTc prolongation),
 - DA antagonists: prochlorperazine, metoclopramide (used in CA chemotherapy and also a prokinetic in GERD as discussed – mentioned above).
 - H₁ antagonists: diphenhydramine, meclizine, promethazine
 - Muscarinic antagonists: scopolamine

Prokinetics

- Drugs used for treatment of GIT motility disorders like achalasia, gastroparesis (delayed GIT emptying), and other forms of dysmotility.
- Include cholinergic agents, dopamine receptor antagonists, serotonin receptor antagonists, motilin agonists and miscellaneous drugs.

a. Cholinergic agents: Bethanechol and neostigmine methylsulfate

b. Dopamine-Receptor Antagonists

- ✓ Dopamine normally inhibits GI motility and can irritate vomiting center (D2R)
- ✓ Its antagonists are therefore, used as prokinetics and antiemetic
- ✓ Includes: **Metoclopramide**, Domperidone (more selective D2 antagonist)
- ✓ **Metoclopramide:** One of the oldest true prokinetic agent with additional mechanisms:
 - 5-HT₄-receptor agonist, Vagal and central 5-HT₃-antagonism, Possible sensitization of muscarinic receptors on smooth muscle
 - Its effect is confined to the **upper GI tract**, no clinically significant effects on **large-bowel motility**.
- ✓ P/K: Absorbed rapidly after oral ingestion; Undergoes sulfation and glucuronide conjugation by the liver; excreted principally in the urine, with a half-life of 4 to 6 hours
- ✓ **Therapeutic use**
 - ✓ Symptomatic treatment in **GERD, gastroparesis**
 - ✓ Its greatest USE lies in its ability to ameliorate the nausea and vomiting that often accompany GI dysmotility syndromes
- **ADR:** EPS (extrapyramidal symptoms); tardive dyskinesia; galactorrhea (excess PLN)

c. Serotonin Receptor Modulators

- ✓ Control motor and secretory function of the gut
- Serotonin Receptors: either **stimulatory** (5-HT₃ & 4- causing motility and secretion) or **inhibitory** (5-HT_{1a})
- Include: Tegaserod Maleate and Cisapride

d. Motilin agonists

- Motilin is a **potent contractile agent** (peptide hormone) of the **upper GI tract**
- Mimicked by Macrolides: oleandomycin, azithromycin, and clarithromycin and Erythromycin

Erythromycin: in diabetic gastroparesis if metoclopramide can't be used.

e. Miscellaneous: Clonidine (alpha₂ agonist)- for **gastroparesis**

Vitamin Supplements

- Are given when concurrent long-term drug therapy depletes vitamin stores or otherwise interferes with the function of a vitamin. For e.g. Vitamin B₆ is given to pts taking INH for TB.
- Can also be given in GI malabsorption syndromes and vitamin deficiencies (pernicious anemia for e.g. – give Vitamin B₁₂)
- Additionally, they can be given for therapeutic purposes at mega doses (e.g. Niacin to lower cholesterol).

- Can be toxic (hypervitaminosis) esp. the fat soluble ones (except vitamin E). Hypervitaminosis is less likely to occur with water soluble vitamins except vitamin B6 (pyridoxine) – which can cause nerve damage at very large doses (when taken 200X the DRI ¹³)
- Vitamin A supplement (Retinol, Retinal) – indicated for infants, pregnant women, for acne, psoriasis; ADR: can elevate intracranial pressure and cause headache; toxicity is more common in children: increased intracranial pressure (bulged fontanel), irritability, drowsiness, etc...
- Vitamin D supplement: calcitriol (1,25-dihydroxy vitamin D) – can decrease absorption if used with oral laxatives and cholestyramine; toxicity...mostly in children – impairment of renal function and osteoporosis, toxicity is related to hypercalcemia (so additional to removal of the supplement; Calcium diet should be restricted).
- Vitamin E supplement (Tocopherol): functions: antioxidant, **should be given to premature infants** (because high risk of deficiency is speculated).
- Vitamin K supplement (phylloquinone): essential for synthesis of blood coagulation factors (vitamin K dependent clotting factors – II, VII, IX and X) in liver; also used to reverse effects of anticoagulants such as warfarin.
- Vitamin B1 supplement: indicated in Beriberi and Wernicke's encephalopathy, Neuritis (in pregnancy)
- Vitamin B2 supplement: indicated in microcytic anemia and migraine headaches
- Vitamin B3 supplement: indicated in pellagra and hyperlipidemia (at high doses);
- Vitamin B6 supplement: Iron deficiency, morning sickness
- Vitamin B12 supplement: Pernicious anemia; for pregnant women (folate also)
- Vitamin C supplement: for prevention of scurvy; also a urine acidifier

¹³ Dietary reference intake (DRI) – represents the optimal nutrients amounts for good health. While RDA (Recommended daily allowance) represents minimum nutrient requirements.